INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY
INVENTORY

Inventory Control System

Structured Systems Group,
Applewood Computers and Vincent B Coen.

Inventory Control System

September 20, 1979 and later

Copyright (c) 1979 - 2025.

Table of Contents

1. How to Use This Manual....4
2. Features and Facilities....5
3. Overview of Set up and Operation....6
4. Practice Run....9
5. Adapting Office Procedure....12
6. Operational Background....15
7. Setting up the Display Management System....18
8. General and Audit Parameters....19
9. Item File and Sort Parameters....20
10. Disk Drive Allocation and Sorting Considerations....21
11. Building the Item File....22
12. Adding and Depleting Stock....23
13. Audibility and Transaction Proofs....25
14. Adding , Deleting, and changing file Items....26
15. Producing the Reports....27
16. Operational Overview....33
17. System Installation....35
18. Interacting with The System....37
19. Setting System Parameters and drive Allocation....38
20. Building the item file....42
21. Adding, Changing, Deleting & Examining file Items....44
22. Sorting the item file....46
23. Recopying the item file....55
24. Adding to Stock....56
25. Depleting Stock....58
26. Producing the Reports....59
27. DMS Display/Screen Formats....60
28. DMS Setup for non-standard CRT's....64
29. Error Messages....69
CBASIC Error Messages....69
Inventory Control System Error Messages....70
30. Error Recovery....80
31. Command Summary....82
32. Inventory Specifications Manual....83
I. Distribution Disk Programs....86
II. File Accessing....87
III. File Descriptions....97

1. How to Use This Manual

For best results read through this manual thoroughly at least once before operating the system and do the practice run outlined in Chapter 4.

The Inventory Control System documentation includes two manuals. This manual is a Users' Manual which provides all the background information and instructions necessary for successful operation. The other is a Specifications Manual which contains technical information on file accessing and file descriptions. The Specifications Manual is available at the rear of this document, see chapter 32.

The Users' Manual divides into an Introduction, a Reference Section, an Operations Section, and an Appendix. The Reference sections deal with general concepts and procedures. The Operations sections give additional specifics on running the programs.

While Structured Systems Group is glad to provide helpful support to registered owners of its products, if you have a problem, please make certain you have read and understood the documentation and checked for hardware malfunctions before giving us a call.

2. Features and Facilities

The Structured Systems Group Inventory Control System provides small to medium businesses with up-to-date and accurate information on the quantity, value, and activity of their inventory items. The Inventory Control System has been designed to serve as the central program module of a complete inventory management system. The present Inventory Control System is the first component of that complete inventory package. It stands able to meet the full needs of many businesses. The system performs all the basic inventory control functions such as adding to and depleting stock, adding new items to stock, deleting items no longer carried, keeping track of quantities on hand, quantities on order, quantities back-ordered, and inventory values. Access to file records by item number is quick, allowing timely management information and fast customer service.

A special "auditability option" lets you create an ongoing hard copy record of stock additions, depletions, or both additions and depletions. If data is accidentally entered incorrectly, or if discrepancies arise between your accounting and inventory records, the hard copy transaction proofs simplify the job of tracking down the errors.

The Inventory Control System generates a number of informative reports, including:

1. A detailed printout of the master item file, called the ITEM LIST, that includes information on vendor number, retail price, average unit value, most recent unit cost, quantity in stock, reorder point, reorder quantity, date ordered, date due, quantity on order, and quantity back-ordered.

2. A STOCK ACTIVITY REPORT that details the current period or period to date activity for all or a selected range of stock items. In addition to current quantity on hand, retail price, unit cost, and stock value by item, the report displays quantities added, quantities depleted, and the net change in stock.

3. A STOCK VALUATION REPORT that gives the average value of the item, the replacement cost, stockroom quantity, stockroom value, retail price for each item, and the total inventory value for all items included in the report.

4. A REORDER REPORT that automatically selects all out-of-stock items, items below the reorder point, or items on order, as well as a range of these items.

See Chapter 15 for sample reports.

The Inventory Control System incorporates SSG's Display Management System (DMS), a full-screen formatting utility. Like all Structured Systems Group packages the Inventory Control System employs the most advanced human engineering techniques. Defensive programming ensures successful system operation and trauma-free learning. Edited input and logical commands make for the clearest and easiest operation possible.

The Inventory Control System permits up to four disk drives and three item files. The maximum number of inventory item records supported by the system is 32,767.

3. Overview of Set up and Operation

Because the Inventory Control System is menu driven, you run the programs by selecting them from the system menu. The Inventory Control System also employs SSG's Display Management System, a screen formatting system that simplifies data and parameter entry. By moving the CRT cursor from field to field within the display you enter or change the values in the fields. The system edits your input, and issues descriptive error messages to ensure smooth operation.

SYSTEM SET UP

The first step in system set up is to make a "system master" and "work disk" from the "distribution disk" received from SSG. The second step in system set up is to define the characteristics of your CRT to the Display Management System. Step three of system setup is to set the system parameters. There are four sets of parameters:

1. General System Parameters: These determine the company name, page length, system date, date format (American or international), and whether to sound the console "bell" if an error occurs.

2. Item File Parameters: These establish the number of item files (up to 3 are allowed), their drive locations, and the starting item number for each item file.

3. Audit File Parameters: These determine whether the auditability option is to be used, and if chosen, establishes whether the audit file is for recording additions or depletions, or both, and which drive location holds the audit file.

4. Sort File Parameters: These establish the input, work file, and output drive locations for sorting the item files.

Finally, step four is to actually build the item file by entering information for each inventory item.

NORMAL OPERATION

Once your system is set up and the item file built, normal operation involves adding quantities received, depleting quantities moved out of stock, and producing the reports.

You can access inventory records from the item list in three ways:

1. By entering the drive name and record number relative to the beginning of the file on that drive (i.e., the 4th record on drive B).

2. By entering the item number.

3. Sequentially, by a next-record, or previous-record command.

Besides entering the quantity to be added to stock, you can also change the unit cost, date ordered, date expected, quantity on order, and quantity back-ordered. If you want it to, the

program automatically recomputes the inventory value for that item by the weighted average method.

To deplete stock you enter the quantity leaving your stock. If you desire, the program automatically recomputes the value of the remaining inventory by subtracting the value of the items leaving stock. This value is determined by multiplying the average value of a stock item by the number of units leaving stock.

If you choose not to calculate the value of your inventory by the weighted average method, yet still wish to keep track of your inventory value, you must calculate and change the inventory value manually.

Should you mistakenly add more units to stock than were actually received, you can fix the error by reversing the transaction. Reversing a transaction is handled by adding a negative amount to stock, not by depleting the stock in the amount of the error. This method of "backing out" a transaction maintains the accuracy of the stock activity report. Erroneous depletions are handled similarly.

If you choose the auditability option, normal operation also involves generating TRANSACTION PROOF REPORTS after each session of adding to stock or depleting stock. The transaction proof reports are a hard copy record of the actual entries to the system, and thus serve as a check on the accuracy of your entries, or a way to trace bookkeeping discrepancies. Each transaction proof is numbered and dated to provide a continuous record of your entries to the system.

At the end of each current or to-date period, the START A NEW ACTIVITY PERIOD program lets you zero out the activity registers.

REPORT PRODUCTION

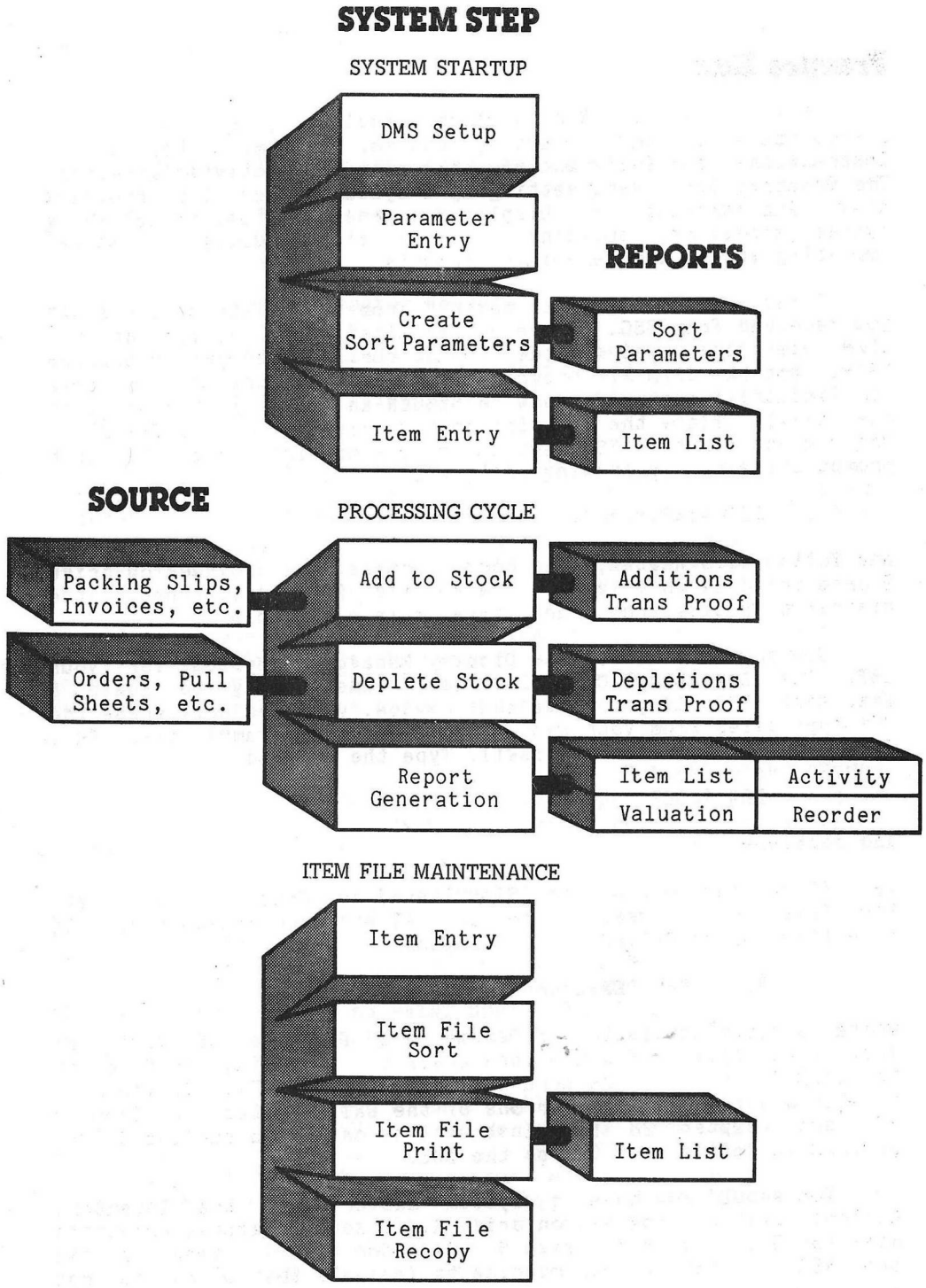
Generating reports is a matter of choosing the report from the system menu and telling the system which report options you want to use (such as reporting only on a range of inventory items for example). Chapter 15 presents examples of the reports available from the Inventory Control System.

FILE MAINTENANCE

When a new item is added to the item file, sorting may be needed, depending on the item number. After a sort, the program requires you to check for duplicate item numbers by running the ITEM FILE PRINT program which can be instructed to print the entire item list with duplicate items flagged, or to print only duplicates if found.

Duplicate file items or items no longer in stock can be removed by a delete command. After deletions are made, the ITEM FILE RECOPY program physically removes the deleted records from the file,thereby compacting it to save disk space.

You can change the item file parameters, for example, to inform the system that additional drives are to be used, or to redistribute the inventory items onto different drives.


SYSTEM STEP
SYSTEM STARTUP
DMS Setup
Parameter Entry
Create Sort Parameters
Item Entry
REPORTS
Sort Parameters
Item List
SOURCE
PROCESSING CYCLE
Packing Slips, Invoices, etc.
Orders, Pull Sheets, etc.
Add to Stock
Deplete Stock
Report Generation
Additions Trans Proof
Depletions Trans Proof
Item List
Activity
Valuation
Reorder
ITEM FILE MAINTENANCE
Item Entry
Item File Sort
Item File Print
Item File Recopy
Item List

4. Practice Run

This Practice Run section provides a quick hands-on introduction to the Inventory Control System. Follow the instructions carefully and make the entries exactly as directed. The Practice Run covers setting up a system master (or program) disk, implementing the Display Management System, establishing system parameters, building an item file, adding to stock, depleting stock, and generating reports.

First, create a "system master" from the "distribution disk" you received from SSG. Place a formatted disk containing the CP/M operating system, the CBASIC runtime interpreter program CRUN2, and the CP/M PIP program on drive A (see the CP/M Features and Facilities manual for how to SYSGEN an operating system onto the disk). Place the SSG distribution disk (write-protected) on drive B and hit the RESET switch on your machine. When the CP/M prompt character is showing (A>), type:

```
PIP A:=B:*.*[OV]
```

and follow with RETURN; this command copies the programs on drive B onto the disk on drive A. When the PIP ends, remove the distribution disk from B and store it in a safe place.

Now you will set up the Display Management System for your CRT. The Inventory Control System comes ready to run on a Hazeltine 1500. If you have that device, you need only erase the DEF type files from your system master (or "program") disk (not from the SSG distribution disk!). Type the command,

```
ERA *.DEF
```

and continue.

If you have one of the CRT's listed in Chapter 17, locate the file that corresponds to your CRT and then rename that file to a file called CRT.DEF by the command,

```
REN A:CRT.DEF=A:nnnnnnnn.ttt
```

where nnnnnnnn.ttt is the file name and file type of your CRT definition file. You may then erase the remaining DEF files by the CP/M ERA command (be sure not to erase CRT.DEF). If you have neither a Hazeltine 1500 nor one of the CRT's listed in Chapter 17, see Chapter 28 for instructions on how to run the CRTDEF program to define your CRT to the DMS.

You should now have a system master with the Inventory Control System programs on drive A. Insert an empty, formatted disk for a data disk on drive B. Give the CTRL-C command or hit the RESET switch on your machine to indicate that a new disk has been inserted. You are now ready to begin operating the system.

To begin, enter:

```
CRUN2 IY
```

at the CP/M prompt character, and follow with RETURN.

Before continuing, the system creates a default parameter file. When it asks you to ENTER SYSTEM DRIVE, enter A followed by RETURN; this is the drive where all parameter files will be located.

First up is the PARAMETER ENTRY MENU (see Chapter 27 for reproductions of the screen formats). At the SELECT A SCREEN request enter 1 followed by RETURN to bring up the GENERAL PARAMETERS screen. Make no changes to the values shown on this screen, except to enter your company name, if desired. The CRT cursor should already be at the company name field.

If you make an error while keying in any entry to the system the command CTRL-H backs the cursor up so you can correct the entry. Enter ESCAPE at any point to go from the General Parameters screen to the Parameter Entry menu; enter ESCAPE once more to continue.

When asked for the date enter 2/1/25. Follow with RETURN. The ITEM FILE PARAMETERS screen which appears next displays the defaults you will use. Make no changes or entries. Enter ESCAPE to go on to the main system menu.

At the SYSTEM MENU select program 9, CREATE SORT PARAMETER FILES, and follow with RETURN. After a message tells which drive holds the file to be sorted (the input drive), give the temporary work file drive as drive B and follow with RETURN. Enter N (for NO) followed by RETURN when asked if you want to switch disks. When asked if you want to PRINT SORT PARAMETERS? answer YES, then RETURN. Turn on your printer, align the paper, and hit RETURN to print.

When the system menu returns, select program 10, ITEM ENTRY, and hit RETURN. When the cursor is in the Item Number field, enter P-1, for item number P-1. Hit RETURN. After the program displays the ACCESS KEY, the cursor moves to the description field. Make entries for the item by filling in the blanks with the information in the table below, hitting RETURN to advance to the next field if a field is not filled. (Use CTRL-B to back the cursor up one field if necessary.) After making all the entries for the item, hit ESCAPE to begin making entries for the second item. Enter P-2 in the item number field and continue as before. After you've entered information for the last item, hit ESCAPE to return the cursor to the item number field. Enter ESCAPE again to move the cursor to the access key field.Enter ESCAPE from there to return to the system menu.

ITEM #	DESC.	STOCK ROOM	REO PT.	REO QTY	RETAIL PRICE	INV. VALUE	VEND. #	COST/ UNIT	QTY ORD	DATE ORD	DATE DUE	B-ORD
P-1	hammer	12	6	18	9.55	60.00	00001	5.00				
P-2	saw	0	5	12	18.99	0.00	00001	10.00	12	1/1/25	3/1/25	0
P-4	nails	20	30	100	1.50	20.00	00002	1.00	100	1/7/25	3/7/25	0
P-3	tacks	15	10	50	1.25	7.50	00002	.50				
P-5	tape	12	15	25	1.75	8.00	00015	.75	25	1/7/25	3/7/25	0
P-6	glue	25	10	35	2.00	25.00	00015	1.00				

At the system menu, select program 11, ITEM FILE SORT. When the sort ends, the system menu reappears. Select program 12, ITEM FILE PRINT. Answer YES to the question, DO YOU WISH TO PRINT ITEM FILES? and follow with RETURN (be sure your printer is ready).

You've now completed the system set up and are ready to begin normal processing. Choose program 1, ADD TO STOCK, from the system menu. When the ADDITIONS TO STOCK screen appears with the cursor in the item number field (see Chapter 27 for a reproduction of the display format), enter the dummy transaction data in the table below. After entering the data for a transaction hit ESCAPE to begin the next. After entering the last addition to stock, hit ESCAPE, and ESCAPE again to return to the system menu.

ITEM #	QTY ADDED	UNIT COST	QTY ORD	DATE ORD	DATE DUE	QTY B-ORD
P-2	12	10.00		NONE	NONE	
P-4	75	1.00				25
P-5	25	1.00		NONE	NONE	

Ready your printer, then select program 3, TRANSACTION PROOF, to print the additions audit file, a record of your entries to the system.

When the system menu returns, select program 2, DEPLETE STOCK. A message that the existing audit file has been proofed, but is not of the depletions type appears. It asks if you wish to erase it. Answer YES, then RETURN. When the DEPLETIONS FROM STOCK screen appears, enter the data in the table below, hitting the RETURN key after entering the quantity depleted to begin the next transaction. After entering the last transaction, hit ESCAPE to return to the system menu.

ITEM #	QUANTITY DEPLETED
P-1	2
P-3	6
P-4	30
P-5	10
P-6	16

Now run program 3, TRANSACTION PROOF, to print the depletions audit file. The system menu returns after the printout.

Print the REORDER REPORT by selecting program 4. When asked to SELECT A REPORT OPTION, key in 3 to indicate you want a report on items that are understocked (whose stockroom quantity is lower than the reorder point). First make sure your printer is turned on, with the paper adjusted. The system menu reappears after the printout.

Print an ACTIVITY REPORT by choosing program 6 from the system menu. When asked to select the ACTIVITY REPORT OPTION, enter 1 to generate a report that covers all items. The system menu returns after the report prints out. Enter ESCAPE from there to end the Inventory Control System. That's all there is to it.

5. Adapting Office Procedure

Adapting your office routine to accommodate computerized inventory control is a minor process in most cases. As with any endeavour, careful planning and forethought smooths the process considerably. This section suggests procedures for setting up, normal operation, using the reports, and backing up for safety.

SET UP

From your company records or a physical inventory compile a list of the items you stock. Arrange the items into logical groupings if possible. Besides the normal operational advantages, such an arrangement means the ACTIVITY, REORDER and VALUATION REPORTS cover related items when you exercise the report range option.

Assign an item number to each item, leaving numbers open between logical groupings to allow for adding more items at a later time. You may use your current numbering system, or renumber the list to conform to the system's 10 character item number limit. A cross-reference between old and new numbers could be maintained, either manually, or automatically with SSG's Analyst system.

Item numbers may contain the characters A through Z, the numbers 0 through 9, slashes (/), and dashes (-). Be sure the slashes and dashes line up. For example, the dashes in the following item numbers are aligned properly:

```
A180-005-6
A180-005-7
A180-016-5
```

To enter the number A180-5-6 instead of A180-005-6 would be improper. In the examples above, the first character of the item number (A) gives the warehouse location. The next character (1) the department number, the next two (80) give the dissection, the three characters following the dash (005 and 016) are a product code, and the final number is a size code.

After compiling the complete list of inventory items and assigning them numbers, make an accurate determination of the quantity on hand and value for each, either by physical inventory or company records.

In addition, you should have retail price, cost per unit, reorder point, and standard reorder quantity information for each item. For items currently on order you should also have information on the date ordered, date due, quantity ordered, and, if applicable, quantity back-ordered.

You may assign a vendor number for each item. The system is set up to use the 5 digit vendor numbers of the SSG Accounts Payable system. If you buy the same item through several vendors, you can establish a separate record for each vendor, or leave the vendor number blank.

NORMAL OPERATION

Under normal circumstances, additions to stock can be keyed directly into the system when they arrive, or batched for later processing. You can take the information from packing slips, invoices, and other original media, or prepare a form of your own. The packing slips and invoices can then be routed through your operation normally. If you choose the auditability option, the batch processing method (saving up additions or depletions for entry all at once) helps ensure the accuracy of your transaction proof reports (see Chapter 13).

You can handle depletions from stock similarly. Enter the depletions when they occur, or batch them for later entry. You can take the information directly from customer orders, pull-sheets, or forms designed especially for the task.

Reorder or backorder information can be entered directly from your purchase order books, buy sheets, or other media.

USING THE REPORTS

The reports generated by the Inventory Control System provide management, accounting, and stockroom personnel with up-to-date and accurate information on availability, turnover, order status, inventory value, as well as retail and replacement costs.

If you choose the system's auditability option, after adding to stock or depleting it you must generate a TRANSACTION PROOF as a check on your entries. Since the TRANSACTION PROOF details either additions or depletions, but not both on the same report, batch processing is recommended if auditability is selected. Batching the entries means you generate one report for additions, and one report for depletions.

The Inventory Control System prepares the reports listed below at any time:

1. ITEM LIST: a multi-purpose listing of all inventory items for use by management, buyers, counter, and stockroom personnel.

2. ACTIVITY REPORT: most useful for the company buyers and marketing staff.

3. STOCK VALUATION REPORT: useful information for the company accountant, controller, or other management. In some states, a real time saver at tax time.

4. REORDER REPORT: a real help to your buyer, and up-to-date information on availability for counter personnel or salespeople.

BACKING UP FOR SAFETY

In order to achieve greater storage, the Inventory Control System cannot automatically back up item files. Therefore, it is important for you to have at least one up-to-date copy of the file at all times. The simplest procedure is to make exact copies of the disks containing the item (or even parameter) files with the full-disk copy program that should have been provided with your disk hardware. If this program is not available, use the PIP command, or the SDCOPY program provided on the SSG distribution disk.

SDCOPY works only for copying single density 8" IBM 3740 standard compatible disks on a CP/M system. To run the program, type SDCOPY at the CP/M prompt, and follow with RETURN. When the message PUT SOURCE ON A, DESTINATION ON B, THEN TYPE RETURN appears, you remove any disks currently in the A and B drives and put the disk to be copied on A. Put an empty, formatted disk (or any disk you don't mind writing over) on B. Hit RETURN to begin. When the copy is complete the reboot message PLACE SYSTEM DISK IN DRIVE A, TYPE RETURN TO REBOOT appears. Remove the two disks and insert the disk that was originally on A. You can use the same disks over and over as the backup disks.

A printout of the ITEM LIST also provides a margin of safety. Should your hardware be out of operation for an extended period, or your files destroyed, this hard copy backup enables you to track the changes in stock manually until the problem is taken care of.

6. Operational Background

Hardware and Software Requirements

The SSG Inventory Control System will run on any 8080 or Z-80 based microcomputer under the control of the CP/M operating system or a CP/M compatible operating system. Your microcomputer system must have the following minimum hardware and software components:

1. 48K (kilobytes) of Random Access Memory (RAM). (CP/M configured for 45K or larger.)

2. Two floppy disk drives with a minimum capacity of 200K.

3. A 132 column wide printer capable of skipping to the top of the next form in response to the ASCII form-feed character.

4. For the console device, a CRT (Cathode Ray Tube) with a direct cursor addressing facility, a clear screen control, and minimum dimensions of 24 lines by 80 columns.

5. System software including an operating system and the CBASIC2 language.

Many microcomputers can be expanded or adapted to meet these requirements. Your local dealer can give you specific information about the hardware currently available.

THE CBASIC PROGRAMMING LANGUAGE

The programs which make up the Inventory Control System are written in a dialect of the BASIC programming language, called CBASIC-2, which is a product of Software Systems. (Questions relating to the CBASIC language should be referred to Software Systems.)

Structured Systems Group distributes its programs in an intermediate file format (with the file type INT), which must be further translated at run time by a facility of the CBASIC language called the "interpreter". The interpreter's program name CRUN2 must precede any INT program for it to run. For example, to run the Inventory Control System you would enter CRUN2 IY. This run time interpreter comes as a part of the CBASIC language package. With the CBASIC language you can also write programs yourself to extend the capabilities of the Inventory Control System, or to perform other data processing tasks.

THE CP/M OPERATING SYSTEM

An operating system is a set of instructions which enables the computer to, among other things, read and write information on floppy disks, move and maintain files, and provide for console communication. The Inventory Control System employs the CP/M operating system, or one compatible with CP/M. CP/M (TM) is a product of the Digital Research Company, and any questions concerning the CP/M system specifically should be referred directly to that company.

CP/M comes documented by a series of six manuals. The first of the six, "Features and Facilities", contains useful and important information relating to your operation of the

Inventory Control System. Be particularly careful to read and understand the DIR, ERA, REN, PIP, SYSGEN, TYPE, and STAT commands before using the Inventory Control System.

LOGGING ON

When you first turn your computer on, its random access memory (RAM) is completely empty. In this state the computer can do nothing. Usually, however, a feature has literally been built into the microcomputer hardware which directs it to look on a predetermined section of a certain disk drive to find what to do next. If everything is working normally, your computer should look on the outer two tracks of the A disk, find the CP/M operating system (if it has been SYSGEN'ed onto the disk; see the CP/M "Features and Facilities" manual for details on how this is done), load it into the RAM, and then display "A>" on the CRT, indicating it is ready to go to work. This process is called "booting" since the computer is, in effect, lifting itself up by its own bootstraps.

The "A>" character is called a "prompt". When the A> prompt appears on the screen, the operating system is said to be "logged on" to the A disk drive. While logged on to the A drive, you can call up programs or data files on that disk by typing the program name. To request a file on another disk (unless this is done automatically by the program you are using) first log on to that disk by keying in the drive name (the letter B on a two disk system), followed by a colon (:) and a carriage return and then type the program name. (The carriage return is simply the RETURN key as it is on a typewriter; it is referred to as CR.)

For example,

```
A>          [system prompt]
A>B:        [user enters B:, then RETURN]
B>
```

To return to the A drive, enter:

```
B>A: cr
```

Whenever the prompt appears you are operating under the CP/M environment. It goes away when you enter the Inventory Control System, and returns when you leave it.

FULL-DISK BACKUP

Caution and common sense dictate that whenever you purchase or receive a new program or system on a floppy disk, you immediately copy it onto a fresh disk and store the original in a fire-proof, moisture-proof box to protect it against inevitable human error and unpredictable system failure. With the original SSG distribution disk tucked safely away from the ravages of dirt, grease, magnet, and mangle, you avoid the cost and inconvenience of unnecessary replacement. This warning applies doubly for disks containing data crucial to your operation. Do not make the mistake of being "penny wise and pound foolish"; the disk itself is almost always worth far less than the information residing on it.

The manufacturer of your disk drive(s) should have provided a full-disk copy program with the disk hardware. If this program is not available, use the PIP command that comes with

the CP/M operating system, or the SDCOPY program provided with the Inventory Control System and explained in the previous chapter.

To usePIP place a fresh disk containing an operating system and the PIP program on drive A, and the Inventory Control System distribution disk on drive B. Hit the reset switch on your machine and then type

```
A>PIP A:=B:*.*[OV]
```

This will copy all the programs and files on B onto the A disk. See the CP/M documentation supplied with your system for a complete description of the PIP program.

CHANGING DISKS BETWEEN PROGRAMS

Whenever you change disks between programs it is of the utmost importance that you enter a CTRL-C character immediately after inserting the new disk. The CTRL-C character, which appears on your screen as ^C, informs the computer that a new disk with different data has been inserted. This "bootstrap" operation (or "re-booting") causes the CP/M system to reset its internal maps of disk usage, and prevents it from writing over (thus destroying) files which exist on the new disk.

To enter CTRL-C, or any CONTROL character, hold down the CONTROL key (CTRL or CTL on some machines), while simultaneously pressing the letter C (either upper or lower case). Never key in a CTRL-C character unless the CP/M prompt is showing. Doing so may end the Inventory Control System, with a probable loss of data.

7. Setting up the Display Management System

The Structured Systems Group Display Management System is a full-screen information display and data entry utility. Chapter 27 of this manual shows each of the screen formats used by the Inventory Control System. Data and parameter entry to the Inventory Control System takes place by "filling in the blanks" on the screen. The Display Management System makes it possible for related information and requests to appear on the screen together. This makes for easy record or parameter review and fast, accurate, data entry.

Because most CRT devices differ, the Display Management System must be informed of the characteristics of your particular device. These characteristics are held in a file of the type DEF that is read by DMS. There are three different methods for defining your CRT to the DMS, depending on your situation:

1. If you have a Hazeltine 1500 device, you need do nothing besides erase all the DEF type files on your system master or program disk (but not the SSG distribution disk!), as well as the program CRTDEF.INT.

2. If your CRT appears on the list below you must carry out a simple REN (rename) command (see Chapter 17), and then erase all the DEF files on your system disk, except the one called CRT.DEF. You may also erase CRTDEF.INT.

ADM 3A
BEEHIVE 150
BEEHIVE 152
BEEHIVE 157
CROMEMCO 3100
DYNABYTE NAKED TERMINAL
SOROC IQ 120
IMSAI VIOC

3. If you have neither a Hazeltine 1500, nor one of the CRT's above, you must run a program provided on your disk called CRTDEF. CRTDEF creates a file by an interactive question an answer process that defines your CRT to the Display Management System. Chapter 28 gives complete instructions.

8. General and Audit Parameters

Parameters can be defined as constant elements or factors within which actions or choices are constrained. Because not all users have the same needs, you can adapt the Inventory Control System to your requirements by changing parameter values, rather than by expensive and complex reprogramming. The Inventory Control System parameters divide into four general types:

1. General Parameters
2. Audit Parameters
3. Item File Parameters
4. Sort Parameters

This chapter covers the first two. Chapter 9 covers the last two.

The General Parameters control such things as the company name, date format (whether American or international), page width, lines to print per page, or whether the console bell sounds when an error occurs. Other parameters classified as general parameters determine what time periods (week, month, quarter) are used for accumulating current period stock activity statistics, and what time periods (month, quarter, year) for accumulating to-date activity statistics. The average valuation parameter lets you decide whether you want the system to calculate the value of your stock by the weighted average method and print a VALUATION REPORT. Another parameter sets an array size according to the amount of random access memory you have; the larger the array, the faster inventory items can be retrieved from the file.

The Audit Parameters let you determine whether you want to keep a hard copy record of your addition and depletion entries to the system as a way to trace accounting or inventory errors, and check the accuracy of your entries. You can order a record of additions to stock, a record of depletions from stock, or both. You can also state on which disk drive to place the audit file which accumulates the entry information.

When you implement the Inventory Control System for the first time, you move directly to the parameter entry screens where you can examine the default parameter values and change them as needed. Later, you set the general and audit parameters by selecting the PARAMETER ENTRY program from the system menu.

9. Item File and Sort Parameters

By changing the Item File and Sort Parameters the Inventory Control System can be set up to accommodate different numbers of disk drives, and different numbers of stock items. If the number of items you stock is greater than the number that fits on a single disk, you will need to break up the item file and spread it over several disks by changing the Item File parameters. And, since access to inventory items requires sorted item files, the Sort Parameters must be set to correspond to the item file arrangement.

What are the item file parameters? The system needs to know how many item files you want. Up to 3 files are possible, with one item file only per disk. For each item file you specify, the Inventory Control System must know its disk drive location. If more than one item file is desired, the system will want to know the lowest (first) item number on the second file, and the lowest (first) on the third file.

At first time start up, the program asks for the Item File parameters automatically, right after you've set the General and Audit parameters. Later, if the number of items you stock grows, or if you wish to redistribute the inventory items to different item files, you can change the Item File parameters through the program on the main system menu called ITEM FILE RECOPY. After you change the parameters, the program creates any new item files, and redistributes the inventory items according to the new parameters.

To sort an item file, the system must know where to locate it (the input drive), where to build the temporary work file it needs for the sort (the work file drive), and where to put the sorted file when finished (the output drive). The sort parameter files contain this information.

If your item files are large, or the number of disk drives available is limited, you may need to remove the input file disk and replace it with another disk to receive the sorted output. The Inventory Control System provides for this possibility; the Sort Parameter Files inform the system that you want this switching to occur. The next chapter describes disk drive allocation and sort procedures in more detail.

If your files increase in size or number, you may need to alter the sort parameters. You can do this by choosing the CREATE SORT PARAMETER FILES program from the system menu. At the end of the CREATE SORT PARAMETER FILES function you can print out the sort parameters for a hard copy reference.

10. Disk Drive Allocation and Sorting Considerations

The disk drive you assign each item file depends on how many disk drives you have, how many item files you have, and how large your item files are. The possible configurations range from one item file on a two drive system, to three item files on a four drive system. Chapter 19 suggests the best arrangement for each possibility.

Files other than item files (such as the audit and INT files) may occupy a significant amount of disk space, and thus should be assigned drive locations with this in mind. The program (INT) files take up the most room, and are always assumed to be on the currently logged drive. The two parameter files (IYPR1 and IYPR2) and the one to three sort parameter files (type SRT) take up little room, and reside on the drive you specify at the beginning of the program as the "system drive". The system drive must be either drive A or drive B. The audit file may be large or small depending on how often you proof and erase the current batch of transactions and begin a new one. You specify the drive location of the audit file at the AUDIT PARAMETERS screen that comes up when you choose the PARAMETER ENTRY program from the system menu.

An item file must be sorted only when you add an item to it out of order or cause it to become disordered by changing an item number. Only the file affected must be sorted. For example, if item file #1 begins with item number K400 and runs to item number M800, adding an item with a number that falls in between (if, of course, such a number is available; otherwise a duplication would occur) such as L200 necessitates a sort. After the sort, you must check for duplicates by running the ITEM FILE PRINT program. Only the ITEM ENTRY, ITEM FILE PRINT, and ITEM FILE SORT programs can be run when unsorted files exist.

Only item file #1 may be sorted from the menu, and then only if disk space is available (on any drive) to hold the temporary work files needed for the sort. There need not be space available for the sorted output, since you can arrange to replace the input (unsorted) disk with another disk to receive the sorted output. The disk used to receive the sorted output may not contain an item file of any kind.

To sort the other item files, or item file #1 if it is large, you end the Inventory Control System and use a special sorting disk (which contains the QSORT program and the appropriate sort parameter files), to run the sort outside the system. A printout of the sort parameters you specified (input location, work file location, and output location for each part file) which you can generate at the end of the CREATE SORT PARAMETER FILES program helps you keep track of where each disk should be. Complete details and a step-by-step explanation are found in Chapter 22.

11. Building the Item File

The item file is a master list of all items in your stock. Each item takes up one record on the file. Each record contains the item number, a description of the item, its cost, its retail price, the value of the quantity on hand, reorder information, vendor number, and information on activity.

Building the item file is the final step in system set up. If no valid records exist on the file, as is the case when you first install your Inventory Control System, the system goes immediately into the adding mode when you choose the ITEM ENTRY program from the menu. Once valid items exist on the file, a control command is needed to add additional records to the file.

To add items to the file, choose the ITEM ENTRY program from the system menu. When the item entry screen (see Chapter 27) appears, enter the adding mode by the appropriate control command, or begin adding if this is the first time set up. Once in the item entry adding mode you enter the item's number. After the program displays the item's drive location and record number relative to the beginning of the file on that drive, you enter a description of the item (for example, "Sport shirts, lg.") and the current stockroom quantity. Enter the reorder point, reorder quantity, vendor number, cost per unit (usually the last price paid), the quantity on order, the date ordered, the date due in, and the quantity back ordered (if any). After making these entries you can move the cursor from field to field indefinitely by RETURN, or begin the next record by hitting ESCAPE.

The ITEM ENTRY program is primarily for building the item file, but can also be used to examine, change, or delete items from the file. A SAVE command is also available; this command updates the disk directory so the entries you make are not lost should the program end prematurely due to a hardware failure (such as a pulled plug).

You can access records to examine or delete them by entering the item number in the ACCESS KEY field, by entering the drive and number of the item relative to the beginning of the file on that drive, or by a next-record or previous-record command.

You may also use the ITEM ENTRY program to update or change the reorder information, the item description, and vendor number. If, however, the stockroom quantity is altered through this program, the stock activity report will not reflect this change, and will thus be inaccurate. If auditability has been requested you should not alter the inventory value through this program since it would not be reflected in the transaction proofs.

12. Adding and Depleting Stock

Once your system is set up and the master item file built, additions to and depletions from stock are handled by the ADD TO STOCK and DEPLETE STOCK menu functions. When you select the ADD TO STOCK or DEPLETE STOCK programs from the system menu the ADDITIONS TO STOCK or DEPLETIONS FROM STOCK screens appear with the CRT cursor in the ITEM NUMBER field (see Chapter 27 for reproductions of the DMS screen formats). You may then call up the appropriate records by one of three methods:

1. Enter the item number.

2. Enter the drive location and record number relative to the beginning of that drive's item file.

3. Give a next-record or previous-record command.

When the record appears on the STOCK ADDITIONS screen you can enter the number of units received and their cost, and update the order information. The program automatically adds to the stockroom quantity for the item, subtracting like amounts from the quantity ordered and back ordered figures (the quantity back-ordered is treated as a subset of the quantity on order). If you've chosen the average valuation feature the program also increases the inventory value by the amount derived from multiplying the number of units times the unit cost. It also adds the number of units to the current period and to-date period activity registers for the item which are the basis of the ACTIVITY REPORT. If you've chosen to audit additions to stock, the transaction is recorded in the audit file.

When items leave your stock you use the DEPLETE STOCK program to enter the change. Access to records works the same as for addition to stock. When the record appears, you enter the number of units leaving your stock. The DEPLETE STOCK program decreases the stockroom quantity figure and adds to the current period and to-date period activity registers.

If you've chosen average valuation the DEPLETE STOCK program also decreases the inventory value for the item based on the weighted average value of the item. For example, if 5 units were added at $10 each, 5 units at $12 each, and 10 units at $20 each the weighted average is figured as follows:

```
 5 @ $10 = $50
 5 @  12 =  60
10 @  20 = 200
   ----  -----
    20    310
```

Weighted average = 310/20 = $15.50 per unit

Should a depletion from stock occur now, the inventory value for the item would decrease at the rate of $15.50 per unit. The weighted average is recomputed each time new items are added to stock.

If you've chosen to audit depletions from stock the transaction is recorded in the audit file. If you've ordered both additions and depletions audited you must print a TRANSACTION PROOF of the current audit file, and then erase it, before you can go from the ADD TO STOCK program to the DEPLETE STOCK program or vice versa. The transaction proof is a hard copy record of your addition and depletion entries and their effect on inventory value. You generate the transaction proof by choosing the TRANSACTION PROOF program from the system menu. With the auditability option, batching transactions (grouping them for entry all at once) is best, although you could move back and forth from addition to depletions as long as you printed the transaction proof and erased the audit file each time. See Chapter 13 for more information on the auditability option and transaction proofs.

Additions and Depletions should only be made through the addition and depletion programs, not through the ITEM ENTRY program. This keeps the activity registers accurate, and records the transactions in the audit file.

If the number of units added to stock is too large, you must correct the mistake by making a negative addition to stock in the amount of the error, not by making an offsetting depletion. Similarly, if the amount depleted is too large, you must "back out" of the error by making a negative depletion, not an offsetting addition. This makes sure the activity registers do not reflect additions and depletions that did not actually take place.

If you choose to track the value of your inventory by the average valuation facility, the amount subtracted from the inventory value (when backing out an addition to stock error), or the amount added (when backing out a depletion error) could differ from the original amount if the average value for the item has changed. Allowing the audit file to contain only add to stock transactions or deplete stock transactions, but not both at the same time, increases the chance that you will catch the error on the on the transaction proof before the average value of the item is altered by an intervening addition at a different cost per unit.

13. Audibility and Transaction Proofs

The audibility option of the Inventory Control System allows the user to leave a hard copy inventory transaction history. In addition to its use in documenting sound financial and managerial practice, this hard copy trail facilitates error tracing and recovery. It provides an ongoing check on the accuracy of your inventory control system, making your automated inventory and your physical inventory match as closely as possible.

The hard copy printout is called a TRANSACTION PROOF. The transaction proof shows the item added or depleted, the quantity, the unit cost, and the change in stock value brought about by the transaction. It also shows addition or depletion errors that were backed out as cancelled transactions.

You can order additions only, depletions only, or both additions and depletions to be audited. You make these choices when setting the system's AUDIT PARAMETERS. A special audit file is created at the beginning of the ADD TO STOCK or DEPLETE STOCK program, and resides on the disk drive you specify in the AUDIT PARAMETERS until erased.

If you begin by adding to stock, the audit file created will be of the additions type and will only accept add to stock transactions. To maintain the accuracy of your transaction history you must print a transaction proof covering all additions transactions and then erase the additions audit file before you choose the DEPLETE STOCK program from the menu. If you try to use the deplete stock program while the additions audit file exists, a message informs you that the current audit file is of the additions type. The message says whether all the transactions in the file have been proofed, and asks if you wish to erase it.

If you begin by making depletions, a depletions audit file is created and likewise must be proofed and erased before you return to the add to stock program.

Of course, you can choose not to proof the audit file, erase it, and go on to the desired function. Doing so, however, means the transaction proofs will not provide a complete and accurate record of your entries to the system, and thus somewhat defeats the purpose of the audibility feature.

Each time an audit file is erased and a new one created the batch number that identifies the audit file increases by one. Each time an audit file is proofed, the proof number for that batch increases by one. Thus you can tell by looking at the audit proofs that a particular audit file was never proofed since a batch number would be missing. However, even if the batch numbers are contiguous the audit trail may not be complete, since a batch may have been proofed at least once, but not after additional transactions have been entered. You should always print a transaction proof before erasing a file.

14. Adding , Deleting, and changing file Items

As you add new items to your stock and remove others no longer carried, you'll want to add or delete item file records. Your may also want to alter the reorder point, reorder quantity, retail price, vendor number, cost per unit, or update order dates and quantities for an item. These additions, deletions, and changes are made through the ITEM ENTRY program.

All records added to the file are placed at the end of the file regardless of the item number of the record. If the item number of the new inventory item is greater than the item number of the last record on the file, the file remains in sorted order. If it is less, adding the record puts the file out of order. The program issues a message that if the item is added the file will need to be sorted. After sorting, to make sure the file does not contain items with duplicate numbers, you must run the ITEM FILE PRINT program. This program prints out the entire item list, marking any duplicates with asterisks, or, if you desire, prints out only duplicate records that it finds.

To add a new item you call the ITEM ENTRY program from the menu and give the command for adding new items. Enter the item number and then the information for the item.

When you first build the item file, you might want to leave certain item numbers unused so new items of the same logical type will be placed on the file at appropriate points. Doing so will mean that when you choose the option of reporting on a range of items, the range will include related items.

To delete an item no longer carried in stock, you call the ITEM ENTRY program from the menu, bring up the record, and issue the delete command. Although the deleted record remains physically on the file until you run the ITEM FILE RECOPY program, you cannot access or report on it.

To change an item on the file, call the ITEM ENTRY program from the menu, bring up the record and change the desired fields. If you have chosen the auditability option, never change the stockroom quantity field through the ITEM ENTRY program; use the ADD TO STOCK or DEPLETE STOCK programs. If you've chosen to use the average valuation feature, make no changes to the inventory value field; it adjusts automatically whenever stock is added or depleted. Changing the item number may put the file out of order and necessitate a sort.

15. Producing the Reports

You can order any of the four reports produced by the Inventory Control System at any time by choosing them from the menu. You can order an ITEM LIST, a REORDER REPORT, a STOCK ACTIVITY REPORT, or a stock VALUATION REPORT.

To print the ITEM LIST choose the ITEM FILE PRINT program from the system menu. You can print the entire list, or only a selected range of items. The ITEM LIST shows the disk drive location, relative record number, item number, item description, vendor number, retail price, inventory value, cost per unit, number in stock, reorder point, reorder quantity, date ordered, and quantity back-ordered. In addition, this report states the total number of records, the total number of deleted records, and the drive location for each item file.

Using the REORDER REPORT you can report on items whose stockroom quantity has fallen below the reorder point, on items out of stock completely, or on items currently on order. For each of these reorder report options you can order a report on all, or only on a range of items.

You can order an ACTIVITY REPORT on all items, on items active in the current period (week, month, or quarter), or on items active during the month, quarter, or year-to-date period. For each activity report option you can choose to report on all, or only a selected range of items. At the end of the current period (week, month, or quarter), or at the end of the to-date period (month, quarter, or year) you must reset the activity registers to zero. To do this you run the START AN ACTIVITY PERIOD program.

The inventory VALUATION REPORT is available whether or not you have chosen to use the average valuation facility. However, if you do not choose average valuation, the report will be meaningful only if you manually compute the changes in inventory value and enter them through the ITEM ENTRY program. You can order the report on all file items, or only a selected range. In addition to giving the inventory value by item, the report also prints a total value for all items.

Representative report samples appear on the following pages.

MARTIN'S MENSWEAR--WAREHOUSE #1

INVENTORY

ITEM FILE PRINT

DATE 10/2/79

DISK DRIVE	RECORD NUMBER	ITEM	VENDOR NUMBER	PRICE	VALUE	COST	NUMBER IN STOCK	REORDER POINT	REORDER QUANTITY	DATE ORDERED	DATE DUE	QTY ON ORDER	QTY ON B/ORDER
B	1	A180-001-5	00005	200.00	15,000.00	100.00	150	25	100	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, wool knit, short												
B	2	A180-001-6	00005	210.00	21,000.00	105.00	200	50	250	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, wool knit, regular												
B	3	A180-001-7	00005	210.00	7,350.00	105.00	70	25	100	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, wool knit, long												
B	4	A180-001-8	00005	220.00	2,750.00	110.00	25	10	40	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, wool knit, X-Long												
B	5	A180-002-5	00005	225.00	2,200.00	110.00	20	25	50	09/01/79	11/01/79	50	0
	DESCRIPTION: Suits, wool knit, vested, sht												
B	6	A180-002-6	00005	230.00	15,000.00	115.00	100	30	150	08/01/79	10/01/79	15	15
	DESCRIPTION: Suits, wool knit, vested, reg.												
B	7	A180-002-7	00005	240.00	3,000.00	120.00	25	30	50	08/01/79	09/15/79	50	50
	DESCRIPTION: Suits, wool knit, vested, long												
B	8	A180-002-8	00005	250.00	1,250.00	125.00	10	15	30	09/01/79	11/01/79	30	0
	DESCRIPTION: Suits, wool knit, vested, X-lg												
B	9	A180-005-6	00014	175.00	20,000.00	100.00	200	50	150	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, Gabardine, regular												
B	10	A180-005-7	00014	175.00	10,000.00	100.00	100	25	75	00/00/00	00/00/00	0	0
	DESCRIPTION: Suits, Gabardine, long												
B	11	A187-016-5	00007	100.00	1,500.00	50.00	30	40	50	09/01/79	11/01/79	60	0
	DESCRIPTION: Sportcoats, wool, short												
B	12	A187-016-6	00007	100.00	2,500.00	50.00	50	40	60	00/00/00	00/00/00	0	0
	DESCRIPTION: Sportcoats, wool, regular												
B	13	A187-016-7	00007	100.00	2,500.00	50.00	50	25	45	00/00/00	00/00/00	0	0
	DESCRIPTION: Sportcoats, wool, long												
B	14	A250-001-1	00025	15.00	0.00	10.00	0	35	72	09/01/79	10/15/79	72	0
	DESCRIPTION: Dress shirts, cotton, small												
B	15	A250-001-2	00025	15.00	2,000.00	10.00	200	100	250	00/00/00	00/00/00	0	0
	DESCRIPTION: Dress shirts, cotton, medium												
B	16	A250-001-3	00025	15.00	1,000.00	10.00	100	50	100	00/00/00	00/00/00	0	0
	DESCRIPTION: Dress shirts, cotton, large												
B	17	A250-023-1	00026	12.00	1,800.00	6.00	300	100	250	00/00/00	00/00/00	0	0
	DESCRIPTION: Dress shirts, polycotton, sm.												

THIS IS A PARTIAL LISTING COVERING ITEM NUMBER A180-001-5 THROUGH ITEM NUMBER A250-023-1

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
R E O R D E R R E P O R T
ITEMS FROM A180-001-5 THROUGH A250-023-1

RECORD KEY	STOCK FLAG	ITEM NUMBER	D E S C R I P T I O N	QTY STKROOM	REORDER POINT	STD REORDER	VENDOR NO.	UNIT COST	ON ORDER	DATE ORDERED	DATE DUE	BACK ORDERED
@B1		A180-001-5	Suits, wool knit, short	150	25	100	00005	100.00	0			0
@B2		A180-001-6	Suits, wool knit, regular	200	50	250	00005	105.00	0			0
@B3		A180-001-7	Suits, wool knit, long	70	25	100	00005	105.00	0			0
@B4		A180-001-8	Suits, wool knit, X-Long	25	10	40	00005	110.00	0			0
@B5	U	A180-002-5	Suits, wool knit, vested, sht	20	25	50	00005	110.00	50	09/01/79	11/01/79	0
@B6		A180-002-6	Suits, wool knit, vested, reg.	100	30	150	00005	115.00	15	08/01/79	10/01/79	15
@B7	U	A180-002-7	Suits, wool knit, vested, long	25	30	50	00005	120.00	50	08/01/79	09/15/79	50
@B8	U	A180-002-8	Suits, wool knit, vested, X-lg	10	15	30	00005	125.00	30	09/01/79	11/01/79	0
@B9		A180-005-6	Suits, Gabardine, regular	200	50	150	00014	100.00	0			0
@B10		A180-005-7	Suits, Gabardine, long	100	25	75	00014	100.00	0			0
@B11	U	A187-016-5	Sportcoats, wool, short	30	40	50	00007	50.00	60	09/01/79	11/01/79	0
@B12		A187-016-6	Sportcoats, wool, regular	50	40	60	00007	50.00	0			0
@B13		A187-016-7	Sportcoats, wool, long	50	25	45	00007	50.00	0			0
@B14	O	A250-001-1	Dress shirts, cotton, small	0	35	72	00025	10.00	72	09/01/79	10/15/79	0
@B15		A250-001-2	Dress shirts, cotton, medium	200	100	250	00025	10.00	0			0
@B16		A250-001-3	Dress shirts, cotton, large	100	50	100	00025	10.00	0			0
@B17		A250-023-1	Dress shirts, polycotton, sm.	300	100	250	00026	6.00	0			0

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
R E O R D E R R E P O R T
ITEMS ON ORDER

RECORD KEY	STOCK FLAG	ITEM NUMBER	D E S C R I P T I O N	QTY STKROOM	REORDER POINT	STD REORDER	VENDOR NO.	UNIT COST	ON ORDER	DATE ORDERED	DATE DUE	BACK ORDERED
@B5	U	A180-002-5	Suits, wool knit, vested, sht	20	25	50	00005	110.00	50	09/01/79	11/01/79	0
@B6		A180-002-6	Suits, wool knit, vested, reg.	100	30	150	00005	115.00	15	08/01/79	10/01/79	15
@B7	U	A180-002-7	Suits, wool knit, vested, long	25	30	50	00005	120.00	50	08/01/79	09/15/79	50
@B8	U	A180-002-8	Suits, wool knit, vested, X-lg	10	15	30	00005	125.00	30	09/01/79	11/01/79	0
@B11	U	A187-016-5	Sportcoats, wool, short	30	40	50	00007	50.00	60	09/01/79	11/01/79	0
@B14	O	A250-001-1	Dress shirts, cotton, small	0	35	72	00025	10.00	72	09/01/79	10/15/79	0

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
R E O R D E R R E P O R T
UNDERSTOCKED ITEMS

RECORD KEY	STOCK FLAG	ITEM NUMBER	D E S C R I P T I O N	QTY STKROOM	REORDER POINT	STD REORDER	VENDOR NO.	UNIT COST	ON ORDER	DATE ORDERED	DATE DUE	BACK ORDERED
@B5	U	A180-002-5	Suits, wool knit, vested, sht	20	25	50	00005	110.00	50	09/01/79	11/01/79	0
@B7	U	A180-002-7	Suits, wool knit, vested, long	25	30	50	00005	120.00	50	08/01/79	09/15/79	50
@B8	U	A180-002-8	Suits, wool knit, vested, X-lg	10	15	30	00005	125.00	30	09/01/79	11/01/79	0
@B11	U	A187-016-5	Sportcoats, wool, short	30	40	50	00007	50.00	60	09/01/79	11/01/79	0
@B14	O	A250-001-1	Dress shirts, cotton, small	0	35	72	00025	10.00	72	09/01/79	10/15/79	0
@B19	U	A250-023-3	Dress shirts, polycotton, lrg.	100	125	100	00026	6.00	0			0

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
D E P L E T I O N S P R O O F
BATCH NUMBER 1, PROOF NUMBER 1 FOR TRANSACTIONS DATED 10/2/79

ITEM NO.	D E S C R I P T I O N	DEPLETIONS QUANTITY	CHANGE IN STOCK VALUE
A187-016-7	Sportcoats, wool, long	30	1,500.00
A250-001-2	Dress shirts, cotton, medium	50	500.00
A250-001-3	Dress shirts, cotton, large	75	750.00
A180-001-5	Suits, wool knit, short	25	2,500.00
A180-001-6	Suits, wool knit, regular	50	5,250.00
A180-001-7	Suits, wool knit, long	25	2,625.00
A180-001-8	Suits, wool knit, X-Long	25	2,750.00
A255-012-1	Sport shirts, cotton, small	75	300.00
A255-012-2	Sport shirts, cotton, medium	100	400.00
A255-012-3	Sport shirts, cotton, large	75	300.00
A255-012-3	Sport shirts, cotton, large	25	100.00

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
A D D I T I O N S P R O O F
BATCH NUMBER 2, PROOF NUMBER 1 FOR TRANSACTIONS DATED 10/2/79

ITEM NO.	D E S C R I P T I O N	ADDITIONS QUANTITY	UNIT COST	CHANGE IN STOCK VALUE
A180-002-5	Suits, wool knit, vested, sht	45	110.00	4,950.00
A180-002-6	Suits, wool knit, vested, reg.	15	115.00	1,725.00
A180-002-7	Suits, wool knit, vested, long	50	120.00	6,000.00
A180-002-8	Suits, wool knit, vested, X-lg	28	125.00	3,500.00

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY
V A L U A T I O N R E P O R T

RECORD NUMBER	ITEM NUMBER	DESCRIPTION	AVERAGE COST	REPLACEMENT COST	STOCKROOM QUANTITY	VALUE IN STOCKROOM	RETAIL PRICE
@B1	A180-001-5	Suits, wool knit, short	100.00	100.00	125	12,500.00	200.00
@B2	A180-001-6	Suits, wool knit, regular	105.00	105.00	150	15,750.00	210.00
@B3	A180-001-7	Suits, wool knit, long	105.00	105.00	45	4,725.00	210.00
@B4	A180-001-8	Suits, wool knit, X-Long	0.00	110.00	0	0.00	220.00
@B5	A180-002-5	Suits, wool knit, vested, sht	110.00	110.00	65	7,150.00	225.00
@B6	A180-002-6	Suits, wool knit, vested, reg.	115.00	115.00	115	13,225.00	230.00
@B7	A180-002-7	Suits, wool knit, vested, long	120.00	120.00	75	9,000.00	240.00
@B8	A180-002-8	Suits, wool knit, vested, X-lg	125.00	125.00	38	4,750.00	250.00
@B9	A180-005-6	Suits, Gabardine, regular	100.00	100.00	200	20,000.00	175.00
@B10	A180-005-7	Suits, Gabardine, long	100.00	100.00	100	10,000.00	175.00
@B11	A187-016-5	Sportcoats, wool, short	50.00	50.00	30	1,500.00	100.00
@B12	A187-016-6	Sportcoats, wool, regular	50.00	50.00	50	2,500.00	100.00
@B13	A187-016-7	Sportcoats, wool, long	50.00	50.00	20	1,000.00	100.00
@B14	A250-001-1	Dress shirts, cotton, small	0.00	10.00	0	0.00	15.00
@B15	A250-001-2	Dress shirts, cotton, medium	10.00	10.00	150	1,500.00	15.00
@B16	A250-001-3	Dress shirts, cotton, large	10.00	10.00	25	250.00	15.00
@B17	A250-023-1	Dress shirts, polycotton, sm.	6.00	6.00	300	1,800.00	12.00
@B18	A250-023-2	Dress shirts, polycotton, med.	6.00	6.00	400	2,400.00	12.00
@B19	A250-023-3	Dress shirts, polycotton, lrg.	6.00	6.00	100	600.00	12.00
@B20	A250-025-1	Dress shirts, polycotton, sm.	7.00	7.00	100	700.00	14.00

TOTAL VALUE: 102,850.00

THIS IS A PARTIAL LISTING COVERING ITEM NUMBER A180-001-5 THROUGH ITEM NUMBER A250-025-1

MARTIN'S MENSWEAR--WAREHOUSE #1
INVENTORY

STOCK ACTIVITY REPORT
ITEM NUMBERS RANGING FROM A180-001-5 THROUGH A250-025-1

REC NO.	ITEM NO.	D E S C R I P T I O N	CURRENT STOCK	RETAIL PRICE	UNIT COST	STOCK VALUE		ADD-ITIONS	DEP-LETIONS	NET CHANGE
B1	A180-001-5	Suits, wool knit, short	125	200.00	100.00	12,500.00	CURRENT MONTH:	0	25	-25
							YEAR TO DATE:	0	25	-25
B2	A180-001-6	Suits, wool knit, regular	150	210.00	105.00	15,750.00	CURRENT MONTH:	0	50	-50
							YEAR TO DATE:	0	50	-50
B3	A180-001-7	Suits, wool knit, long	45	210.00	105.00	4,725.00	CURRENT MONTH:	0	25	-25
							YEAR TO DATE:	0	25	-25
B4	A180-001-8	Suits, wool knit, X-Long	0	220.00	110.00	0.00	CURRENT MONTH:	0	25	-25
							YEAR TO DATE:	0	25	-25
B5	A180-002-5	Suits, wool knit, vested, sht	65	225.00	110.00	7,150.00	CURRENT MONTH:	45	0	45
							YEAR TO DATE:	45	0	45
B6	A180-002-6	Suits, wool knit, vested, reg.	115	230.00	115.00	13,225.00	CURRENT MONTH:	15	0	15
							YEAR TO DATE:	15	0	15
B7	A180-002-7	Suits, wool knit, vested, long	75	240.00	120.00	9,000.00	CURRENT MONTH:	50	0	50
							YEAR TO DATE:	50	0	50
B8	A180-002-8	Suits, wool knit, vested, X-lg	38	250.00	125.00	4,750.00	CURRENT MONTH:	28	0	28
							YEAR TO DATE:	28	0	28
B9	A180-005-6	Suits, Gabardine, regular	200	175.00	100.00	20,000.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B10	A180-005-7	Suits, Gabardine, long	100	175.00	100.00	10,000.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B11	A187-016-5	Sportcoats, wool, short	30	100.00	50.00	1,500.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B12	A187-016-6	Sportcoats, wool, regular	50	100.00	50.00	2,500.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B13	A187-016-7	Sportcoats, wool, long	20	100.00	50.00	1,000.00	CURRENT MONTH:	0	30	-30
							YEAR TO DATE:	0	30	-30
B14	A250-001-1	Dress shirts, cotton, small	0	15.00	10.00	0.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B15	A250-001-2	Dress shirts, cotton, medium	150	15.00	10.00	1,500.00	CURRENT MONTH:	0	50	-50
							YEAR TO DATE:	0	50	-50
B16	A250-001-3	Dress shirts, cotton, large	25	15.00	10.00	250.00	CURRENT MONTH:	0	75	-75
							YEAR TO DATE:	0	75	-75
B17	A250-023-1	Dress shirts, polycotton, sm.	300	12.00	6.00	1,800.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B18	A250-023-2	Dress shirts, polycotton, med.	400	12.00	6.00	2,400.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B19	A250-023-3	Dress shirts, polycotton, lrg.	100	12.00	6.00	600.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0
B20	A250-025-1	Dress shirts, polycotton, sm.	100	14.00	7.00	700.00	CURRENT MONTH:	0	0	0
							YEAR TO DATE:	0	0	0

16. Operational Overview

SET UP

System set up involves four steps:

1. Make two copies of the SSG distribution disk.
2. Define your CRT to the Display Management System.
3. Set the system parameters.
4. Build the item file.

To make the copies of the distribution disk you need two formatted disks with the CP/M operating system SYSGEN'ed on (see the CP/M Features and Facilities manual on how to do this). These disks should also hold the CP/M PIP and STAT commands, as well as the CBASIC runtime interpreter CRUN2. After you PIP all the Inventory Control System files onto your disks, store the SSG distribution disk safely away; use one of your newly created disks as a system master and one as a work disk. If you need to make new or additional work disks, make them from the system master disk, not the SSG distribution disk.

The Display Management System comes ready to run on a Hazeltine 1500 CRT device; if you have a Hazeltine 1500 you just erase the DEF type files and the CRTDEF.INT program from your two disks (not the SSG distribution disk!) and continue. If you have one of the CRT's listed in the following chapter the procedure involves renaming the file that corresponds to your CRT to CRT.DEF; you then erase the rest of the DEF files and the CRTDEF.INT program and begin. If you have neither a Hazeltine 1500, nor any of the CRT's listed in the next chapter, you define your CRT to the Display Management System by running the CRTDEF.INT program and answering its requests.

To set the system parameters you call up the Inventory Control System by the command CRUN2 IY, set the GENERAL PARAMETERS according to your needs, set the AUDIT PARAMETERS, set the ITEM FILE PARAMETERS, and the SORT PARAMETERS. At first time start-up the system takes you immediately to the PARAMETER ENTRY program where you set the general and audit parameters, and then to the section of the ITEM FILE RECOPY program where you set the item file parameters. Once these are set, the system menu comes up and you should choose the CREATE SORT PARAMETER FILES program.

To build the item file you call the ITEM ENTRY program through the system menu. If no valid records exist on the file the program goes right into the adding mode; if records do exist on the file you must give the add records command (CTRL-A). Enter the item number, then the information for the the item. Enter ESCAPE at any time to to place the information in the file and begin adding the next item.

NORMAL PROCESSING

Normal processing includes adding to stock when goods arrive, depleting stock when they leave, generating the reports, and, if you wish auditability, printing the transaction proofs.

To add to stock you select the ADD TO STOCK program from the menu, access the item by item number, relative record number, or sequentially by a next- record or previous

record command, and enter the quantity added and its cost. The program updates the quantities on order and back-ordered by subtracting the amount added from both values. You can then change the order dates or adjust the other order information. ESCAPE enters the transaction and begins the next additions to stock. ESCAPE once more returns you to the system menu.

To record depletions from stock select the menu program DEPLETE STOCK. Access the part as for adding to stock and enter the quantity depleted. The program automatically updates the quantity in stock.

If you've chosen the auditability option (at the AUDIT PARAMETERS screen of the PARAMETER ENTRY program), and want both additions and depletions audited, you must erase the current audit file before going from add to stock to deplete stock, or from deplete stock to add to stock. This is because the audit file holds additions or depletions, but not both kinds of transactions at the same time. To ensure accuracy and completeness you should print a transaction proof before you erase the audit file. If you've chosen to audit additions only, or depletions only, there is no need to erase the file, but you should proof and erase it regularly so it does not grow unnecessarily large.

To generate a report, select the desired reporting program from the system menu, choose the report options available, set up your printer, and print.

When you add new items to your stock they must be added to the master item file through the ITEM ENTRY program. If the addition of the item means the file is no longer ordered by item number, a sort will be required. How you run the sort depends on how many item files you have, how many disk drives you have, and the size of the item files. Chapter 22 gives full details.

If you wish to change the distribution of the items among the item files, or when you want to physically remove deleted records from the file, you run the program ITEM FILE RECOPY from the system menu, changing disks as directed by the program.

If the number of item files changes, or if you want to alter your sorting procedures, you will need to change the sort parameters through the CREATE SORT PARAMETER FILES program.

17. System Installation

The first step in installing the Inventory Control System is to make two system disks from the SSG distribution disk.

Place a freshly formatted disk containing the CP/M operating system, the CBASIC runtime interpreter CRUN2, and the CP/M PIP and STAT commands on drive A. Place the (write protected) SSG distribution disk on drive B and hit the RESET switch on your machine. Type the command,

A>PIP A:=B:*.*[OV]

and follow with RETURN. When the PIP is complete, remove the distribution disk from drive B and replace it with a formatted disk with the CP/M operating system SYSGEN'ed onto it. Reset the machine, then give the command,

A>PIP B:=A:*.*[OV]

The second step in installing the Inventory Control System is to define the characteristics of your CRT to the Display Management System.

If you have the Hazeltine 1500 CRT, the Inventory Control System will run immediately, just erase all files with the file type (or suffix) of DEF and the CRTDEF.INT program from your work disks.

If you have any of the following CRT's you must use the CP/M REN command to rename the DEF file corresponding to your CRT to the filename CRT.DEF:

CRT	DEF FILE
ADM 3A	ADM3A.DEF
BEEHIVE 150	B150.DEF
BEEHIVE 152	B152.DEF
BEEHIVE 157	B157.DEF
CROMEMCO 3100	CRO3100.DEF
DYNABYTE NAKED TERMINAL	DYNABYTE.DEF
SOROC IQ 120	SOROC120.DEF
IMSAI VIOC	VIOC.DEF

Type the command,

A>REN A:CRT.DEF=A:nnnnnnnn.ttt[v]

where nnnnnnnn.ttt stands for the file name and file type of your CRT definition file. Erase the remaining DEF files from your work disk (not the SSG distribution disk!), but do not erase the CRT.DEF file. Erase also the CRTDEF.INT program.

If you have neither the Hazeltine 1500 terminal, nor any of the CRT's listed above, see Chapter 28 for instructions on how to define your CRT to the DMS by running the CRTDEF.INT program.

The Inventory Control System will always expect to find the CRT.DEF file on the currently logged drive.

You can now call up the Inventory Control System and begin setting the system parameters and building the item file, the last steps remaining in the system set up sequence.

18. Interacting with The System

At first time start up the Inventory Control System creates a default parameter file, goes directly into the PARAMETER ENTRY program, then into the ITEM FILE RECOPY program (to set the item file parameters), and then to the system menu. Once the system is installed, the command

A>CRUN2 IY

brings up the system menu immediately. All functions except for sorting under certain circumstances (see Chapter 22), can be performed by choosing programs from the system menu (see Chapter 27 for a reproduction of the system menu).

Entry programs use formatted screen displays with prompts for information and bracketed fields to receive the data or parameter values you enter. RETURN moves the CRT cursor forward along a predetermined route through the fields on the screen. CTRL-B backs the cursor up one field in the DMS screen.

If your entry does not fill the field completely you must hit RETURN to go on to the next field. Whatever you type into the field becomes the current value for that field, regardless of what other characters may still remain in the field. For example, if the current value in the Company Name field is [Ajax Enterprises], and your new entry is "Ajax Inc.", the field would look like this before you hit the RETURN key: [Ajax Inc.rprises]. After hitting RETURN the field would show [Ajax Inc.----------].

To end any Inventory Control System function press the ESCAPE key. ESCAPE takes you one "layer" out of the program. For example, while adding items to the file you hit ESCAPE to begin adding the next item. ESCAPE again takes you out of the adding mode but leaves you at the ITEM ENTRY screen where you can examine, change, or delete other file records. ESCAPE once more removes you from the ITEM ENTRY screen and program altogether and returns the system menu. ESCAPE at the system menu ends the Inventory Control System. The procedure for leaving the ADD TO STOCK and DEPLETE STOCK programs is similar, though you need not hit ESCAPE between depletion transactions. Typing the word STOP works similarly at non-DMS controlled requests.

To cancel a transaction such as entering a new item into the file, adding to stock, or depleting stock, issue the command CTRL-X at any field on the DMS screen. Should a DMS screen become garbled for any reason, the command CTRL-R refreshes it.

To back up one request when not in a Display Management System screen (such as at the report option requests) you should enter an up-arrow (^) followed by RETURN.

To back the cursor up one space in a DMS screen field, use the command CTRL-H. At non-DMS requests, use the DELETE command.

19. Setting System Parameters and drive Allocation

The procedure for setting system parameters at initial start up differs from the procedure for setting the parameters once the system is installed. When no parameter files or valid inventory records exist, after asking for the "system drive", the system goes immediately to the PARAMETER ENTRY program where you can examine or change the general and audit parameters. After these are set, the system moves on to the section of the ITEM FILE RECOPY program that allows you to set or change the item file parameters (see Chapter 23). After the item file parameters are set, the system menu comes up. You should choose the CREATE SORT PARAMETER FILES program, and then the ITEM ENTRY program. If no valid records exist on the item file the program goes automatically into the adding mode, with the cursor in the item number field. You then enter the item number and other information for that item.

After first time start up, to change the parameters you call the appropriate program from the system menu directly. All fields for parameter (and data) entry are edited to exclude double quotation marks (") and invalid control characters, and to ensure the correctness of the entry (e.g., alphabetic characters not allowed in the vendor number field).

GENERAL PARAMETERS

Set the general parameters through the menu function PARAMETER ENTRY. This brings up the PARAMETER ENTRY MENU. Choose GENERAL PARAMETERS and make the desired entries or changes subject to the restrictions listed below:

COMPANY NAME: Maximum length is 60 characters

CURRENT PERIOD: Accepts week, month, or quarter only.

TO DATE PERIOD: Accepts month, quarter, or year only.

DATE FORMAT: Accepts MM/DD/YY or DD/MM/YY.

PAGE WIDTH: Accepts any value from 1 to 999. Since the Inventory Control System requires a 132 column printer, the PAGE WIDTH parameter control the placement of the report headings. A value less than 132 moves the heading to the left, a value greater than 132 moves it to the right.

LINES PER PAGE: Accepts any value from 3 to 999. The default is 60 lines, or 10 inches, measured from the top of the first printed line.

AVERAGE VALUATION: When set to NO, the value of the inventory must be calculated manually, outside the system, and the INVENTORY VALUE field updated through the ITEM ENTRY program if you wish a valuation report. ARRAY SIZE: This value depends on the amount of Random Access Memory (RAM) available. The larger the array, the faster the record access, although unless you expect to access many items during a run of the system a larger array size does not substantially speed up record access. Use the chart below to determine the right size for your computer (memory sizes are CP/M configured):

RAM	ARRAY SIZE
45K	100
48K	175
56K	500
64K	1000

CONSOLE BELL: Sounds the console "bell" when errors occur.

DEBUGGING: Leave set to NO.

AUDIT PARAMETERS

Set the audit parameters through the PARAMETER ENTRY program. When the PARAMETER ENTRY MENU comes up, choose AUDIT PARAMETERS. Enter the disk drive where the audit file is to reside (A-D), and then set the ADDITIONS AUDIT USED and DEPLETIONS AUDIT USED values as desired. For no auditability, set both to NO.

ITEM FILE PARAMETERS

Set or change the item file parameters through the menu program ITEM FILE RECOPY. When asked if you wish to change the parameters answer YES. A NO answer tells the program you don't wish to change the parameters, but want to remove deleted records from the file. If you anticipate only one item file, enter the drive location (A-D). If more than one, enter the drive location for each, and the lowest item number on item file #2 and item file #3. Item numbers may contain only the letters A through Z, the numbers 0 through 9, dashes, or slashes.

SORT PARAMETERS

For each item file you order, there is a corresponding sort parameter file. If you change the number of item files, or if they grow to a size that makes the current set of sort parameters unworkable, you will need to change the values in the sort parameter files. After calling the menu program CREATE SORT FILE PARAMETERS, set the work file and output drive locations, and indicate whether you want to be able to switch disks to receive the sorted output. The program assumes the input drive is the same drive as that specified for the item file. The sort parameter files are named (if three item files exist) PART1.SRT, PART2.SRT, and PART3.SRT.

DISK DRIVE ALLOCATION

The Inventory Control System always expects to find the program files (those with the file type of INT) and QSORT on the currently logged drive. The two parameter files IYPR1.101 AND IYPR2.101 are placed on the drive specified as the "system" drive at system start up. The system drive must be either drive A or B only. If they are found to be on a different drive than the one specified as the system drive, the program assumes that that drive is now the system drive. The sort parameter files PART1.SRT, PART2.SRT, and PART3.SRT are also expected to be on the system drive, unless they are placed on a special sorting disk (see Chapter 22). The audit file drive is established through the PARAMETER ENTRY program and can be on any drive location. The CRT.DEF file (if used) is expected to reside on the currently logged drive.

The chart below suggests the best file allocation for one to three item files on two to four disk drives:

2 DRIVE SYSTEM

```
---------------------------------------------------------------
1 Item File:   A: INT FILES
               QSORT
               SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               B: ITEM FILE #1 (IYPRT)

2 Item Files:  A: INT FILES
               QSORT
               ITEM FILE #1 (IYPRT)
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               ITEM FILE #2 (IYPRT)
```

3 DRIVE SYSTEM

```
---------------------------------------------------------------
1 Item File:   A: INT FILES
               QSORT
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               C: ITEM FILE #1 (IYPRT)

2 Item Files:  A: INT FILES
               QSORT
               ITEM FILE #1 (IYPRT)
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               B: ITEM FILE #1 (IYPRT)
               C: ITEM FILE #2 (IYPRT)

3 Item Files:  A: INT FILES
               QSORT
               ITEM FILE #1 (IYPRT)
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               ITEM FILE #2 (IYPRT)
               C: ITEM FILE #3 (IYPRT)
```

```
4 DRIVE SYSTEM
-------------------------------------------------------------
1 Item File:   A: INT FILES
               QSORT
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               C: ITEM FILE #1 (IYPRT)
               D: not used, or AUDIT FILE

2 Item Files:  A: INT FILES
               QSORT
               B: SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               C: ITEM FILE #1 (IYPRT)
               D: ITEM FILE #2 (IYPRT)

3 Item Files:  A: INT FILES
               QSORT
               SYSTEM DRIVE: IYPR1
                       IYPR2
                       SRT FILES
               AUDIT FILE (IYAUD)
               B: ITEM FILE #1 (IYPRT)
               C: ITEM FILE #2 (IYPRT)
               D: ITEM FILE #3 (IYPRT)
```

20. Building the item file

Once the system parameters have been set, to build the item file, select the ITEM ENTRY program from the system menu. If no valid records are on the item file, the program goes automatically into the adding mode with the cursor waiting at the item number field. If there are valid records on the item file the cursor goes to the access key field; the command CTRL-A moves the cursor to the item number field and puts the program in the adding mode.

To add an item to the file, give the item number and hit RETURN. The item number may not be more than 10 characters long, may contain only the letters A through Z, the numbers 0 through 9, dashes, or slashes. After the program displays the relative record number of the item in the access key field, enter the rest of the information for the item. To advance the cursor after each entry hit RETURN (if the field is not filled), to move back one field use the command CTRL-B, to back up to make an erasure use a CTRL-H command, and to cancel a transaction (such as an item entry), use the command CTRL-X.

Note the allowed entries for each field:

```
DESCRIPTION:      30 characters maximum
STOCKROOM QTY:    999,999
REORDER POINT:    999,999
REORDER QTY:      999,999
RETAIL PRICE:     99,999.99
INVENTORY VALUE:  9,999,999.99
VENDOR NUMBER:    any 5 digit number
COST/UNIT:        99,999.99
QTY ON ORDER:     999,999
DATE ORDERED:     MM/DD/YY or DD/MM/YY
DATE DUE:         MM/DD/YY or DD/MM/YY
BACKORDER QTY:    999,999
```

If you wish the ACTIVITY REPORT to reflect original additions to the file, do not make entries for the stockroom quantity, inventory value (unless you have chosen not to use average valuation), or cost per unit. Instead, make all other entries for the item, then use the ADD TO STOCK program to enter the stockroom quantity, inventory value, and cost per unit.

When you enter an item out of order, the item file is renamed to a file type with a zero in the first position (e.g., IYPRT.001) that indicates it is unsorted. Before you can operate any programs other than ITEM ENTRY, ITEM FILE PRINT, or ITEM FILE SORT, you must sort the file (see Chapter 22).

Once you enter a valid item number, ESCAPE at any point adds the record to the end of the file and readies the program for the next item. Another ESCAPE right after the first takes you out of the adding mode and moves the cursor to the access key field where you can issue commands to access records, delete records, or save records to disk (see Chapter 21). ESCAPE from the access key field returns you to the system menu.

While entering items onto the item file, the CTRL-S command updates the disk directory, thereby saving the records you have created in case a catastrophic hardware failure

occurs that causes the data to be lost (such as a pulled plug). Whenever you have entered more records than you would care to enter again, leave the adding mode and give the command CTRL-S. One unpleasant experience is usually enough to demonstrate the value of this function.

21. Adding, Changing, Deleting & Examining file Items

Use the ITEM ENTRY program to add new items to the file, to change field values for file items, to delete items no longer stocked, or to examine items on the file.

To add a new item select ITEM ENTRY from the system menu, and if valid records exist on the file, issue the CTRL-A command to add records. See Chapter 20 for how to add an item to the file. If adding the item to the end of the file puts the file out of order (for example, if you add item number A2-4600 to an item file whose last record holds item number A3-4550) the file will have to be sorted. A disordered file is renamed to indicate that it is out of order (from IYPRT.101 to IYPRT.001 for example).

When you add a new item to the file and want the stockroom quantity to be reflected on the ACTIVITY REPORT or audit proofs, do not enter that quantity through ITEM ENTRY; instead, enter that information through the ADD TO STOCK program. If you've chosen inventory valuation, or want the addition reflected on the audit proof, do not enter the stockroom quantity, unit cost, or inventory value through ITEM ENTRY, but rather through the ADD TO STOCK program.

To change an item already on the file, access the record by one of three methods when the cursor is at the access key field:

1. Enter the item number.

2. Enter the record number relative to the beginning of the file, in the form @Dnnn, where D is the drive reference, and nnn is the relative record number of the item on that drive. For example, @C25 gets the twenty-fifth record on the item file on drive C.

3. Enter the CTRL-N command to bring up the first record on the item file, CTRL-N for the next record, and CTRL-B for the previous record.

Move the cursor forward to the desired field with RETURN, or back a field with CTRL-B (once a record is accessed). Type in the new value and hit RETURN to enter it. If you want auditability, do not change the stockroom quantity, inventory value, or cost per unit fields through the ITEM ENTRY program. If you choose the average valuation feature, do not change the stockroom quantity, inventory value, or cost per unit through ITEM ENTRY. You may alter the item number, but note that this may put the file out of order and necessitate a sort before you can use any program besides ITEM ENTRY, ITEM FILE PRINT, or ITEM FILE SORT.

To delete an item from the file, access the record as explained above for changing a record and issue the command CTRL-D. Although not physically removed from the file, the item can no longer be accessed. To physically remove deleted records you must run the ITEM FILE RECOPY program (see Chapter 23). Because the record for each inventory item contains more information than is shown on the ITEM ENTRY screen, you should delete rather than change an item (since you cannot change all the information on the record).

To examine an inventory item, call the ITEM ENTRY program from the system menu, and with the cursor at the access key field call the item by item number, relative record number,

or the next-record and previous-record commands. If you have chosen to audit additions and depletions, it is usually best to use ITEM ENTRY to examine the item; besides the additional information, the audit file does not need to be proofed and erased, for example, if it is of the depletions type and you want information from the additions screen.

22. Sorting the item file

The following assumes you are running a system using floppy disks and not a hard drive:

Sorting procedures vary depending on the number of item files, the size of the item files, and the number of disk drives available. This chapter suggests sorting procedures for each of the configurations suggested in Chapter 19. Other procedures besides those given in this chapter are available; you may want to experiment some to find the one best for you.

Only item file #1 can be sorted from the system menu, and then only if there is space available on any drive other than the input drive to hold the work file. The work files created (and erased at the end of the sort) equal the size of the input file. Space to receive the sorted output need not be available since you can instruct the system to prompt you to remove the disk with the original input file and insert an empty disk to receive the output file.

If space is not available for the work file (use the CP/M STAT command; see the CP/M Features and Facilities manual) you need to create a special sorting disk that contains the program QSORT.COM and the necessary sort parameter files. When you sort the item files outside of the Inventory Control System this special sorting disk holds the temporary work files. The input file should not be so large that it cannot fit on the sorting disk with QSORT and the small sort parameter files.

All item files have the same file name of IYPRT. To determine whether an IYPRT file is the first, second, or third item file, use the CP/M TYPE command. The first record to appear is the "header record". The header record contains the file's drive reference and the highest item number on that file.

All item files have a three digit file type (or suffix). A zero as the leftmost digit indicates the file is unsorted, a 1 (one) indicates it is sorted. A 1 (one) as the middle digit indicates the file has been sorted, but not checked for duplicates (if, of course, the first digit is 1). Under normal circumstances, the rightmost digit appears as 1.

SUGGESTED SORTING PROCEDURES

1. One item file on a 2 drive system (with file less than 1/3 disk):

```
A       B

INT FILES          ITEM FILE #1
QSORT.COM
PARAM FILES
SRT FILES
AUDIT FILE
```

Sort Parameters:

```
INPUT DRIVE:      B
OUTPUT DRIVE:     B
WORK FILE DRIVE:  B
SWITCH?:          NO
```

Suggested Procedure: Run directly from menu when sorting required. The (unsorted) input file is erased if the ITEM ENTRY program is run at any time after the sort.

2. One item file on a 2 drive system (with space on a for the work file and the file no larger than 1/2 disk):

A	B
INT FILES	ITEM FILE #1
QSORT	
PARAM FILES	
SRT FILE	
AUDIT FILES	

Sort Parameters:

INPUT DRIVE:	B
OUTPUT DRIVE:	B
WORKFILE DRIVE:	A
SWITCH?:	NO

Sorting Procedure: Run directly from the menu when required. The input file is erased if the ITEM ENTRY program is run at any time after the sort.

3. One item file on a 2 drive system (with item file larger than 1/2 disk & insufficient space on A for the work file):

A	B
INT FILES	ITEM FILE #1
QSORT	
PARAM FILES	
SRT FILES	
AUDIT FILE	

Sort Parameters:

INPUT DRIVE:	B
OUTPUT DRIVE:	B
WORK FILE DRIVE:	A
SWITCH?:	YES

Sorting Procedures: Must run this sort outside the Inventory Control System. You will need to make up a special sorting disk with QSORT.COM, the sort parameter file PART1.SRT, and the CP/M operating system. Perform the following steps:

1. To make the special sorting disk, end the Inventory Control System and remove the disk currently on drive B. Insert an empty, formatted disk (with the CP/M operating system) on B. Give the CTRL-C to reset the maps of disk usage. Type the command PIP

B:=A:QSORT.COM[V], the command PIP B:=A:*.SRT[V], and the command PIP B:=A:PIP.COM[V] (for convenience). The B disk is now your special sorting disk.

2. Remove the disk on drive A and replace with the sorting disk on B. Put the disk that was originally on B on drive B. Give the CTRL-C command. The A drive should now hold the following files: QSORT.COM, PART1.SRT, and PIP.COM. The B drive should hold only the first item file, IYPRT.001 (the file type 001 indicates the item file is unsorted; when sorted it changes to type 101).

3. Type the command QSORT PART1. When the sort finishes, the system prompts you to remove the B disk with the unsorted item file, and replace it with an empty disk to receive the sorted output from the work files on A. Do not reboot.

4. Two item files on a 2 drive system:

```
   A            B

INT FILES       ITEM FILE #2
QSORT           PARAM FILES
ITEM FILE #1    SRT FILES
                AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:      A
OUTPUT DRIVE:     A
WORK FILE DRIVE:  B
SWITCH?:          YES
```

Sort Parameters for Item File #2:

```
INPUT DRIVE:      B
OUTPUT DRIVE:     B
WORK FILE DRIVE:  A
SWITCH?:          YES
```

Sorting Procedure: Item file #1 can be sorted from the menu, without disk switching if there is room on A to receive the sorted output, and room on B to hold the temporary work files (which are the same size as the input file). Otherwise, sort both files outside the Inventory Control System using the parameters given above.

1. Prepare a special sorting disk containing QSORT, PIP, and the two SRT files (and the CP/M operating system). In this case you need to get QSORT from the disk that was on A, and the SRT files from the disk that was on B (the "system drive").

2. Put the sorting disk on B, and the disk with item file #1 on A. Give the CTRL-C command. Type the command B:QSORT PART1 (to run the sort from drive B). Switch the A disk to receive the sorted output when prompted.

3. Remove the disk now in A, set it aside, and insert the special sorting disk that was on B into A. Put the disk with item file #2 on drive B.

Give the CTRL-C command. Type the command QSORT PART2. When instructed by the program remove the B disk and replace it with a disk to receive the sorted output from A.

4. You now have the two original disks which contain the unsorted item files (and other files), one sorting disk, and two disks containing only the sorted files. You must now either PIP the files that were on the original disks onto the new, sorted, disks, or erase the unsorted files from the old disks and PIP the sorted files back in their place. Make sure the files are allocated properly before re-entering the Inventory Control System.

5. One item file on a 3 drive system:

```
   A            B                C

INT FILES   PARAM FILES     ITEM FILE #1
QSORT       SRT FILES
            AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:       C
OUTPUT DRIVE:      C
WORK FILE DRIVE:   B
SWITCH?:       YES (if file is greater than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sorting Procedure: May be run from the system menu. Switch disks when prompted if switching requested. If the parameter files, and audit file on drive B occupy too much space and do not allow the work file to fit, you should PIP those files onto the A drive and erase them from the B drive. Or, you could prepare the special sorting disk as explained above and run the sort outside the system.

6. Two item files on a 3 drive system (with space available on a at least equal to the size of item file #1):

```
   A                 B             C

INT FILES       ITEM FILE #1   ITEM FILE #2
QSORT
PARAM FILES
SRT PARAMS
AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:       B
OUTPUT DRIVE:      B
WORK FILE DRIVE:   A
SWITCH?:       YES (if file is larger than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sort Parameters for Item File #2:

```
INPUT DRIVE:      C
OUTPUT DRIVE:     C
WORK FILE DRIVE:    B
SWITCH?:       YES (if file is greater than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sorting Procedure: May sort item file #1 directly from the system menu. Be sure QSORT is on the currently logged drive (which in this case should be drive A), and PART1.SRT is on the drive named as the system drive (also drive A for this configuration).

1. Select the SORT ITEM FILE program from the menu and switch disks when prompted.
.
2. Sort Item File #2 outside of the system. Follow the procedure for sorting item file #2 as explained in the section below.

7. Two item files on a 3 drive system (with insufficient space available on drive a for the item file #1 work file):

```
   A              B                C

INT FILES      ITEM FILE #1      ITEM FILE #2
QSORT
PARAM FILES
SRT FILES
AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:      B
OUTPUT DRIVE:     B
WORK FILE DRIVE:  C
SWITCH?:      YES (if file is larger than 1/2 disk)
              NO  (if file is less than 1/2 disk)
```

Sort Parameter for Item File #2:

```
INPUT DRIVE:       C
OUTPUT DRIVE:      C
WORK FILE DRIVE:   B
SWITCH?:    YES (if file is larger than 1/2 disk)
            NO  (if file is less than 1/2 disk)
```

Sorting Procedure: Must sort both item files outside the system. Be sure QSORT is on the currently logged drive, and the sort parameter files on the drive named as the system drive (these should both be drive A under this configuration).

1. With QSORT, PART1.SRT, and PART2.SRT on drive A, put the disk with item file #1 on drive B. Put an empty, formatted, disk on drive C. Give the CTRL-C command.

2. Type QSORT PART1. When the sort routine finishes, switch the B disks to receive the sorted output if switching was ordered.

3. Remove the disk now in B and replace with the blank disk that served as the work file on C. Put the disk with item file #2 on drive C and give the CTRL-C command.

4. Type QSORT PART2. Switch disks on drive C to receive the sorted output if switching was ordered. Remove the blank disk from drive B, replace it with the sorted item file #1, give the CTRL-C command, and you're ready to re-enter the Inventory Control System.

8. Three item files on a 3 drive system:

```
   A           B             C

INT FILES     ITEM FILE #2   ITEM FILE #3
QSORT         PARAM FILES
ITEM FILE #1  SRT FILES
              AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:      A
OUTPUT DRIVE:     A
WORK FILE DRIVE:  B
SWITCH?: YES (if space on A less than item file #1)
         NO  (if space on A greater than item file #1)
```

Sort Parameters for Item File #2:

```
INPUT DRIVE:      B
OUTPUT DRIVE:     B
WORK FILE DRIVE:  A
SWITCH?: YES (if space on B less than item file #2)
     NO  (if space on B greater than item file #2)
```

Sort Parameters for Item File #3:

```
INPUT DRIVE:      C
OUTPUT DRIVE:     C
WORK FILE DRIVE:  A
SWITCH?: YES (if space on C less than item file #3)
         NO  (if space on C greater than item file #3)
```

Sorting Procedure: All files must be sorted outside the system.

1. Make a special sorting disk that contains the CP/M system, QSORT, PIP, and the three sort parameter files PART1.SRT, PART2.SRT, and PART3.SRT Place this disk on drive B. Put the disk with item file #1 on drive A and give the CTRL-C command.

2. Type B:QSORT PART1. Switch A disks to receive the sorted output from the work file on B if switching has been requested.

3. To sort item file #2, put the disk that contains that file on drive B, the sorting disk on A, and give the CTRL-C command. Type QSORT PART2. Switch B disks to receive the sorted output from the work file on A if switching has been requested.

4. To sort item file #3, put the disk that contains that item file on drive C, the sorting disk on drive A, and give the CTRL-C command. Type QSORT PART3. Switch C disks to receive the sorted output from the work file on A if switching has been requested.

5. You now have the original A, B, and C disk which contain the unsorted item files and the other Inventory Control System programs and files, one special sorting disk, and three disks containing only the sorted item files. To get back to the original configuration, you must either PIP the programs and other files onto the disks with the sorted files, or erase the unsorted files from the original disks, and PIP the sorted files onto the original disks in their place. Be sure all programs and files reside on the proper disks before restarting the Inventory Control System.

9. One item file on a 4 drive system

```
   A          B              C              D

INT FILES   PARAM FILES   ITEM FILE #1  not used, or
QSORT       SRT FILES                   AUDIT FILE
            AUDIT FILE
```

Sort Parameter for Item File #1:

```
INPUT DRIVE:       C
OUTPUT DRIVE:      C
WORK FILE DRIVE:   A, B, or D
SWITCH?:       YES (if file is larger than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sorting Procedure: Run directly from the system menu. Be sure QSORT is on the currently logged drive and the SRT files on the drive named as the system drive (in this case, B).

10. Two item files on a 4 drive system

```
   A          B              C              D

INT FILES   PARAM FILES    ITEM FILE #1    ITEM FILE #2
QSORT       SRT FILES
            AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:       C
OUTPUT DRIVE:      C
WORK FILE DRIVE:   D
SWITCH?:    YES (if file is larger than 1/2 disk)
            NO  (if file is less than 1/2 disk)
```

Sort Parameters for Item File #2:

```
INPUT DRIVE:       D
OUTPUT DRIVE:      D
WORK FILE DRIVE:   C
SWITCH?:     YES (if file is larger than 1/2 disk)
             NO  (if file is less than 1/2 disk)
```

Sorting Procedure:

1. To sort item file #1, be sure that QSORT is on the currently logged drive (in this case drive A) and the sort parameter files on the system drive (in this case drive B). Put the disk with item file #1 on drive C, and a empty, formatted, disk on drive D. Give the CTRL-C command. Type QSORT PART1. Switch the C disks when prompted if switching has been requested.

2. To sort item file #2, make sure QSORT is on the currently logged drive (A) and the sort parameter (SRT) files on the system drive (B). Put an empty, formatted, disk on drive B, and the unsorted item file #2 on drive D. Give the CTRL-C command. Type QSORT PART2. Switch D disks to receive the sorted output from the work file if switching has been requested.

Make sure all disks and files are in their correct locations before beginning.

11. Three item files on a 4 drive system:

```
   A            B             C                D

INT FILES   ITEM FILE #1 ITEM FILE #2    ITEM FILE #3
QSORT
PARAM FILES
SRT FILES
AUDIT FILE
```

Sort Parameters for Item File #1:

```
INPUT DRIVE:       B
OUTPUT DRIVE:      B
WORK FILE DRIVE:   A
SWITCH?:       YES (if file larger than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sort Parameters for Item File #2:

```
INPUT DRIVE:       C
OUTPUT DRIVE:      C
WORK FILE DRIVE:   A
SWITCH?:       YES (if file larger than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sort Parameters for Item File #3:

```
INPUT DRIVE:       D
OUTPUT DRIVE:      D
WORK FILE DRIVE:   A
SWITCH?:       YES (if file larger than 1/2 disk)
               NO  (if file is less than 1/2 disk)
```

Sorting Procedure:

1. Make up a special sorting disk containing the CP/M system, QSORT, PIP, and the three sort parameter files (SRT).

2. Put the special sorting disk in drive A, the disk with item file #1 on drive B, the disk with item file #2 on drive C, and the disk with item file #3 on drive D. Give the CTRL-C command.

3. Type QSORT PART1. Switch B disks when prompted if switching requested.

4. Type QSORT PART2. Switch C disks when prompted if switching requested.

5. Type QSORT PART3. Switch D disks when prompted if switching requested.

6. Replace the sorting disk in A with the disk containing the other programs and files. Give the CTRL-C command before restarting the Inventory Control System.

23. Recopying the item file

When you alter the item file parameters (see Chapter 19), or when you want to remove deleted records from the file, select the ITEM FILE RECOPY program from the system menu. The program first asks if you want to change the item file parameters. YES allows you to do so. NO tells the program you just want to remove deleted records.

Under the direction and prompting of the ITEM FILE RECOPY program, each item file is recopied, in turn, from the A drive to the B drive. When you are instructed, put the disk with the item file to be recopied on drive A and a new disk on B. Hit RETURN to begin the recopy.

The program tells you when to remove the disk from the A drive and replace it with the disk containing the next item file to be recopied. The program also prompts you to replace the disk on the B drive with another disk at the proper time. The disk you place on B should never contain another item file (a file with the file name of IYPRT).

Since the disks with the item files being recopied often contain other files as well, these files must be moved to the new disks at some point before you can operate the Inventory Control System. This is best done before starting the recopy program, by using the PIP command. Be sure there is room enough on each disk to receive the recopied files. For example, you would not want to increase the size of an item file (by changing the item file parameters) so much that it does not fit on the disk with the other files. If this happens, you should re-allocate the Inventory Control System programs and files (see Chapter 19 for suggestions).

After all the item files have been recopied (you must recopy all of them when running the recopy program), the program checks to see if your system disk is in the drive you designated as the system drive. The system disk is the disk with the two parameter files (IYPR1 AND IYPR2) and the sort parameter files, not necessarily the disk with the INT files. If it does not find the system disk where it should be, the program instructs you to put it there. After checking for the system disk, the program looks on the currently logged drive for the disk with the INT files, and prompts you to put it there if it is not already.

24. Adding to Stock

Use the ADD TO STOCK program to increase the stockroom quantity, or change order information for an inventory item.

To use this program, the item files must be sorted by item number (see Chapter 22), and the audit file must be of the additions type if you have requested additions transactions to be audited. If the audit file is not an additions audit file, you should print a transaction proof of the current depletions file, and erase it, before you begin the add to stock transactions. The ADD TO STOCK program creates a new additions audit file and increases the "batch number" by one.

When the program asks if you want to erase the audit file, you may respond with Y (YES), which erases it, with N (NO), which leaves it intact and ends the ADD TO STOCK program, or with ESCAPE, which is the same as NO.

To make an addition to stock, first call up the record for that inventory item. With the cursor at the ITEM NUMBER field,

1. Enter the item number in its exact form.

2. Enter an @ (at-sign) followed by the drive reference and record number relative to the beginning of the item file on that drive. For example, to get the 36th record on item file #2 located on drive B, enter "@B36" followed by RETURN. The @ (at-sign) indicates that what follows it is a relative record number, and not an item number.

3. Give the CTRL-N command to bring up the first record on the first item file, and after that, to bring up the record following whichever record is present. CTRL-B brings up the record immediately preceding the record currently displayed.

Enter the quantity added to stock and hit RETURN to advance to the next field if the quantity added field is not filled. Enter the cost per unit of the stock received and update the order information fields. Do not enter commas, the program inserts them where needed. If an order is filled and you wish to clear the date fields, type "none" in the field. You may cycle through the fields with RETURN and CTRL-B (for going back a field), adjusting the information as necessary until it is correct. The record itself, and the associated accumulators are not updated until you hit ESCAPE. You may then enter another addition to stock transaction, or enter ESCAPE again to return to the system menu.

An addition to stock increases the stockroom quantity and decreases the quantity on order and the quantity back-ordered. The quantity back-ordered is treated as a subset of the quantity on order.

To back out (correct) an addition to stock that was made too large, enter a negative number in the amount of the error (at the same unit cost as the original entry). Do not correct such an error by an offsetting depletion from stock as this will make the activity report inaccurate. For example, if you discover you have made an entry too large by 10 units, enter -10 in the quantity added field and the appropriate value in the cost per unit

field to correct the error. If the entry was made too small, just add the additional units in the regular way.

25. Depleting Stock

Use the DEPLETE STOCK program to decrease the stockroom quantity when goods leave your inventory.

To use this program, the item files must be sorted by item number (see Chapter 22), and the audit file must be of the depletions type if you have requested depletions transactions to be audited. If the audit file is not a depletions audit file, you should print a transaction proof of the current additions file, and erase it, before you begin the deplete stock transactions. The DEPLETE STOCK program creates a new depletions audit file and increases the "batch number" by one.

When the program asks if you want to erase the sudit file you may respond with Y (YES), which erases it, with N (NO), which ends the DEPLETE STOCK program, or with ESCAPE, which is the same as NO.

To make a depletion from stock, first call up the record for that inventory item. With the cursor at the item number field,

1. Enter the item number in the exact form.

2. Enter an @ (at-sign) followed by the drive reference and record number relative to the beginning of the item file on that drive. For example, to get the 36th record on item file #2 located on drive B, enter "@B36" followed by RETURN. The @ (at-sign) indicates that what follows it is a relative record number, and not an item number.

3. Give the CTRL-N command to bring up the first record on the first item file, and after that, to bring up the record following whichever record is present. CTRL-B brings up the record immediately preceding the record currently displayed.

Enter the quantity leaving stock and follow with RETURN. The program updates the stockroom quantity, and, if average valuation has been requested, decreases the inventory value based on the current average value the item. The cursor returns to the item number field where you can enter a new number to begin the next transaction. ESCAPE returns you to the system menu.

To back out (correct) a depletion made too large, enter a negative amount in the quantity depleted field in the amount of the error. Do not correct the error by making an addition to stock, as this makes the activity report inaccurate.

If you've chosen the average valuation feature, an addition to stock between the time a depletion error is made and the time it is discovered may change the average value of that item. Consequently, the adjustment of the error corrects the stockroom quantity, but the new inventory value, computed using a different average value for the item, may not be the same. The difference will usually not be critical, however, since inventory value is an estimate based on averages anyway. Allowing only an addition or depletion audit file to exist at one time helps prevent this situation from arising, since a careful examination of the transaction proofs may allow you to catch the error before an addition to stock changes the average value of the inventory item.
.he

26. Producing the Reports

Once system set up is complete you can print Inventory Control System reports from the system menu at any time. For all but the ITEM FILE PRINT, the item files must be in sorted order.

To print a report, select the desired report program from the system menu and choose the report options you wish to employ. Be sure your printer is ready and the paper aligned. After the printout finishes, the system menu returns.

For certain reports, a range option is available that lets you select a range of high and low item numbers. Leaving the range values blank by hitting RETURN reports on all inventory items. Giving a low value but not a high value reports on all items with numbers above and including that value. Giving a high value but not a low value reports on all items with numbers below and including that value.

If the average valuation option has not been chosen, the VALUATION REPORT will not be useful unless you have maintained the inventory value manually.

At the end of each current period or to-date period, reset the appropriate activity registers to zero by running the START AN ACTIVITY PERIOD program and answering its requests.

27. DMS Display/Screen Formats

```
                Your Company Name

      I N V E N T O R Y   C O N T R O L   S Y S T E M

[  /  /  ]       PROCESSING CYCLE

[ ]  1 ADD TO STOCK             [ ]  4 REORDER REPORT
[ ]  2 DEPLETE STOCK            [ ]  5 VALUATION REPORT
[ ]  3 TRANSACTION PROOF        [ ]  6 ACTIVITY REPORT
                                [ ]  7 START ACTIVITY PERIOD

                  SYSTEM STARTUP

[ ]  8 PARAMETER ENTRY          [ ]  9 CREATE SORT PARAMETER FILES

              ITEM FILE MAINTENANCE

[ ] 10 ITEM ENTRY               [ ] 12 ITEM FILE PRINT
[ ] 11 ITEM FILE SORT           [ ] 13 ITEM FILE RECOPY

              SELECT A PROGRAM > [  ]
```

```
*********************************************************************

                 Your Company Name
      00/00/00    P A R A M E T E R   E N T R Y   M E N U

             1  GENERAL PARAMETERS
             2  AUDIT PARAMETERS

             SELECT A SCREEN > [  ]
```

Your Company Name

PARAMETER ENTRY

00/00/00 G E N E R A L P A R A M E T E R S

COMPANY NAME [ ]

CURRENT PERIOD [ ] AVERAGE VALUATION [ ]
(WEEK, MONTH, QUARTER)

TO DATE PERIOD [ ] ARRAY SIZE [ ]
(MONTH, QUARTER, YEAR)

DATE FORM [ ] CONSOLE BELL USED [ ]
(1 MM/DD/YY 2 DD/MM/YY)

PAGE WIDTH [ ] DEBUGGING [ ]

LINES PER PAGE [ ]

Your Company Name

PARAMETER ENTRY

00/00/00 A U D I T P A R A M E T E R S

AUDIT DRIVE [ ]

ADDITIONS AUDIT USED [ ]
DEPLETIONS AUDIT USED [ ]

****

Your Company Name

DATE [/ /] INVENTORY CONTROL SYSTEM

I T E M E N T R Y

ACCESS KEY
[ ]

ITEM NUMBER
[ ]

DESCRIPTION [ ]

STOCKROOM QUANTITY [ ]

VENDOR NUMBER [ ]

REORDER POINT [ ]

COST / UNIT [ ]

STD REORDER QTY [ ]

QTY ORDERED [ ]

RETAIL PRICE [ ]

DATE ORDERED [ ]

INVENTORY VALUE [ ]

DATE DUE [ ]

BACK ORDERED [ ]

Your Company Name
INVENTORY CONTROL SYSTEM

DATE [/ /] ADDITIONS TO STOCK BATCH NUMBER [ ]

ITEM NUMBER [ ] QUANTITY ORDERED [ ]

QUANTITY ADDED [ ] DATE ORDERED [ ]

UNIT COST [ ] DATE DUE [ ]

STOCKROOM QUANTITY [ ] QUANTITY BACKORDERED [ ]

DESCRIPTION [ ]

****

Your Company Name
INVENTORY CONTROL SYSTEM

DATE [/ /] DEPLETIONS FROM STOCK BATCH NUMBER [ ]

ITEM NUMBER [ ]

QUANTITY DEPLETED [ ]

STOCKROOM QUANTITY [ ]

DESCRIPTION [ ]

28. DMS Setup for non-standard CRT's

To create a new CRT.DEF file, run the CRTDEF program, answering all the requests that pertain to the CRT you are using. When the program requests the name of the CRT definition file to create, respond with CRT. The program will create a file named CRT.DEF. This file should be placed on your program disk (which should be on the currently logged drive). All of the other DEF files can be omitted from your work disks or system master (but not from the SSG distribution disk), as well as the CRTDEF.INT program.

The utility program CRTDEF is used to define the physical characteristics of a CRT display device (e.g., number of rows and columns, screen clear sequence, cursor addressing sequence, etc.). The program will prompt for values for all CRT control sequences which are supported by the DMS system.

The program should only have to be executed once to define the CRT you are using. It does not have to be kept on the system disk after it is run.

The CRTDEF program does not use screen formatting and will work on any serial CRT. It will request data for almost every conceivable type of CRT control procedure. Not all of these procedures are need by the Inventory Control System programs. The only control sequences that are necessary to run are the direct cursor addressing control characters and translation tables, destructive backspace, and the clear screen character. Any other requests can be ignored by pressing RETURN.

Some control items will cause the system to execute faster because it can take advantage of more sophisticated hardware features.

If you are modifying an existing CRT definition file, the CRTDEF program will present a menu so that you may choose which section of the program to run. Enter the number of your desired selection. If you are creating a CRT definition file for the first time, the menu does not appear and all of the requests that are not applicable can be answered to with a RETURN.

When the program issues a request the current value of the control sequence is displayed and the user is prompted for either a new value or acceptance of the current value. To accept the current value, just press the RETURN key.

When the program starts up, it will request the name of the CRT physical characteristics definition file. For the Inventory Control System, always enter CRT. This is the name of the definition file that is created by CRTDEF and the DEF file the Inventory Control System will look for.

CRTDEF will then request whether this run will create the initial version of the file or edit an existing copy. The first time you run this program enter a C for create. If you are modifying a DEF file that was entered incorrectly, enter an M for modify.

THE CONTROL SEQUENCE REQUESTS:

Most of the CRT control sequences are entered in the form:

<number of items>,<item-1>,<item-2>, . . .

The values of the items are the data characters which will be transmitted to the CRT device to effect the specified control operation. These values are specified as decimal integers between 0-255. To enter an ESCAPE, for example, use the decimal equivalent of the ASCII value which is 27 in decimal. If a control sequence consisted of an ESCAPE followed by a left parenthesis, you would enter:

2,27,40

Where the 2 indicates that there are two characters that must be sent to the CRT for this function and the 27 and 40 are the decimal representations of an ESCAPE and a left parenthesis.

Please note that the CRTDEF program does not check the values for validity. Valid values range from 0 to 255 but any integer value may be supplied. The value modulo 256 will be used.

The values specified are stored in character strings and are written to the disk file as such. CBASIC and CP/M impose restrictions on the values which may be contained in these strings. The restrictions are:

CP/M: CTRL-Z (26) is used as an end-of-file marker.

CBASIC: A RETURN (10) or a LINE-FEED (13) character will denote an end of line sequence.

The way around the restrictions for most terminals which might happen to use these characters for CRT control is to specify the value as <value>+128. This sets the ASCII parity bit to ON. For example, to send a decimal 10, send 138 instead.

The ASCII terminals will ignore the parity bit because the defined ASCII set is 7 bits. (Normally the CBIOS [operating system] will force the parity bit to a fixed value before transmitting it to the terminal, so the problem never really arises.

REQUESTS ISSUED BY CRTDEF:

NUMBER OF ROWS FOR DISPLAY:
This is the number of rows (lines) of vertical display on your CRT.

NUMBER OF COLUMNS FOR CRT DISPLAY:
This is the number of columns (characters across) displayed on your CRT.

SPECIFY CURSOR ROW ADDRESS POSITIONING CONTROL CHARACTERS:
row 1 current value: 0
new value:

Enter the value required by the CRT to position the cursor at the desired row. Don't include the position control character sequence, merely the number that will be translated into a row address. This request will be issued once for every possible row.

SPECIFY CURSOR COLUMN ADDRESS POSITIONING CONTROL CHARACTERS:

```
column 1   current value:  0
           new value:
```

Enter the value required by the CRT to position the cursor at the desired column. Don't include the position control character sequence, merely the number that will be translated into a column address. This request will be issued once for every possible column.

CRT CONTROL: SET CURSOR ADDRESS:
As described above, this request starts with "CRT control:" and thus requires the number of elements to precede the values of the elements and for them to be separated by commas.

Enter the values only if your CRT uses one contrl character sequence to send the cursor to both row and column. Enter the characters required to send the cursor to a ROW and a COLUMN at the same time.

SET ROW ADDRESS PREFIX:
If the CRT requires different control characters for ROW and COLUMN, enter the row characters here (the previous request can be left a zero). If the CRT address row and column at the same time this can be left zero.

SET COLUMN ADDRESS PREFIX:
If the CRT requires different control characters for ROW and COLUMN, enter the column characters here. If the CRT addresses row and column at the same time this can be left zero.

SPECIFY ORDER OF PRESENTATION OF ROW AND COLUMN ADDRESS:
Enter an "R" or a "C" for the order in which the control characters are to be sent.

SPECIFY FORMAT OF ROW AND COLUMN CURSOR CONTROL CHARACTERS:
Most CRT's require control information to be sent as hexadecimal characters, but some require ASCII numbers. Enter an "H" or a "D" for the correct format.

CLEAR SCREEN:
Enter the characters for clearing the CRT screen. This is a required entry.

CLEAR FOREGROUND DATA FIELDS:
If your CRT supports different display intensities, enter the foreground/background control characters for this and the next 2 requests.

ENTER FOREGROUND MODE:

ENTER BACKGROUND MODE:

RETURN CURSOR TO HOME POSITION:

MOVE CURSOR UP ONE ROW:

MOVE CURSOR DOWN ONE ROW:

MOVE CURSOR LEFT ONE COLUMN:

MOVE CURSOR RIGHT ONE COLUMN:

DESTRUCTIVE BACKSPACE:
This sequence must be entered. This is the sequence of characters that will erase the previously typed in character. This will almost always be a backspace followed by a blank followed by a backspace. In decimal, this translates to an 8, a 32, another 8. Key in: 3,8,32,8.

SOUND CONSOLE ALARM
This is normally a 1,7.

ERASE TO END OF CURRENT LINE:

ERASE TO END OF SCREEN:

INSERT A NEW LINE INTO THE DISPLAY:

DELETE CURRENT LINE FROM DISPLAY:

NUMBER OF STORAGE LOCATIONS TO BE SET AT CRT INITIALIZATION:
Specify number of storage locations to be set or press RETURN. If your CRT requires memory to be initialized before operating, enter the number of bytes that must be set at this request.

CURRENT STORAGE ENTRY AND VALUES: 0 0

SET STORAGE ENTRY 1:
Specify memory location and value. This request will be issued for each location specified in the previous request. Enter the decimal value for each byte.

INITIALIZATION SUBROUTINE ADDRESS:
If a machine code routine resides in memory that must be branched to on start up, enter its address in decimal.

INITIALIZATION I/O PORT:
If data needs to be transmitted to an I/O port, enter its number.

DATA TO BE TRANSMITTED TO CONSOLE:
If data must be sent to the console device itself at start up time, enter the number of bytes and their decimal values here.

The requests which follow, or options 8 and 10 on the CRTDEF menu are provided to allow you to change the normal definition of the special function control keys in the system. If, for example, your operating system or computer does not allow free use of a CONTROL-A character (enter adding mode), the CRTDEF program would allow you to define a CONTROL-Y entered at the keyboard to be interpreted by the program as though you had entered a CONTROL-A. Because the facility is completely generalized, it should

be possible to compensate for the special requirements of all of the CP/M derivative operating systems.

In addition, the use of control keys can be completely bypassed if necessary. A prefixing character can be defined that substitutes for a control key. In this manner, for example, an at-sign (@) or an up arrow (^) can be designated as the prefix character. Then an @A sequence would be interpreted the same as a CONTROL-A.

The option #8 requests allow a control character translation table to be entered. It issues up to 31 requests for each control character possible. If, for example, your system has forced you to not use a CONTROL-A (ASCII value of 1) and you will substitute a CONTROL-Y (ASCII value of 25), when the request:

```
CHARACTER  25       CURRENT TRANSLATION   25
ENTER NEW VALUE (OR STOP) >
```

is issued, respond with the value 1. Don't forget that 8 is the backspace and 13 is the carriage return.

The option #10 requests are used only if you cannot use any control characters at all. They will define instead a prefix character and the individual values represented by the prefix character followed by one of the 127 possible ASCII values.

The first request issued is for the value of the prefix character itself. Enter its decimal ASCII value. An at-sign, for example, would be 64. Remember that once a character is defined as the prefixing character it cannot be used as a valid data character anywhere else. This is very important, so choose carefully.

The next request is used merely to quickly fill all 127 elements of the table with a default. The default should be some invalid character combination such as a zero.

The program will then issue up to 127 requests for translation of individual characters. If you wish for the character sequence "prefix-A" (65). Don't forget to enter the same interpreted value at the lower-case value of "A" (97).

If you use the prefix character facility of option 10, you should use the control character translation facility of option 8 to "mask" out all special function control characters. This involves setting them to some harmless but invalid value, such as a zero or 10 (LINE FEED). All programs will then reject this character with an explicit error message.

After all requests have been answered (when the menu is not used) or after each menu function (when the menu is used), the program will display:

```
Specify S(ave), E(dit), or Q(uit)
```

If all the data was entered correctly, enter an S. If the data needs to be edited, enter an E and the menu will appear, allowing the data just entered to be modified. If you are unsure about the entries appropriate for your device, see your dealer.

29. Error Messages

CBASIC Error Messages

SSG software is designed to trap all errors which can arise under normal circumstances, but due to language limitations and/or hardware problems, some errors may occur which the programs themselves cannot handle. These errors are CBASIC errors. They appear in the following form,

ERROR aa

where aa is two alphabetic characters issued by the CBASIC runtime package CRUN2. The occurance of a CBASIC error message usually indicates a serious problem in your INT or data files.

ERROR RE: This is caused by a record in a file which has too few or too many file delimiter characters (quotes or commas), or whose length is incorrect. When a program then attempts to read that record, what is on the disk does not match the record description, and CRUN2 issues this message.

It is possible for this problem to occur because of various hardware problems such as a bad write, a voltage surge, etc., but far and away the most common cause is a bad RAM location, which either changes some character into a quote or a comma, or changes one of these file delimiters into some other character. This error is virtually never the result of a program bug. Once a bad record is in the file, it must be repaired or removed with the editor, or the entire file must be rebuilt. Bad records can be detected by TYPE'ing the file and manually searching for a missing or extra quote or comma. If you are not familiar with the use of the editor, read the CP/M documentation before attempting to change the file. Also read the File Definition in the Inventory Control System Specifications Manual, because some files may require that the header record be changed if a record is manually deleted.

ERROR IV: This error occurs if you try to run a program compiled by CBASIC version 1 with CRUN2, or if the INT file that you are trying to run is null. If the latter is the case, recopy the file from the original disk, using the verify option [vo] of PIP.

ERROR DZ: This error occurs if you try to divide by zero, or if you try to run a program compiled under CBASIC2 with CRUN version 1. Since none of the Inventory Control System programs will allow a division with a denominator of zero, you should verify that your runtime package is indeed CRUN2 (ideally version number 2.04 or higher).

ERROR OM or GARBAGE ON SCREEN DURING AN ENTRY PROGRAM: This means that the program has run out of memory--should not occur if you have at least 48K of RAM and a standard-sized version of CP/M.

Inventory Control System Error Messages

The terms "part" and "item" are used interchangeably throughout the documentation. Messages that instruct you to contact your distributor generally indicate an error condition that should not arise under normal circumstances.

SYSTEM WIDE MESSAGES

IY000 PROGRAM MUST BE RUN FROM MENU: To enter the Inventory Control System, type CRUN2 IY.

IY002 INVALID DRIVE SPECIFIED: Must be a single alphabetic character.

IY005 INVALID DATE: MM/DD/YY or DD/MM/YY only.

IY044 INVALID COMMAND SEQUENCE: Contact your distributor.

IY045 INVALID COMMAND SEQUENCE: Contact your distributor.

IY086 NO VALID PART RECORDS: Build the item file through the ITEM ENTRY menu function.

ITEM ENTRY (PARTENT.INT)

IY101 PART FILE FOUND FROM ABORTED RUN: A part file was found whose type had a final digit of 0. This occurs if a program which writes to a part file ends abnormally. Determine the reason, rename the part file (or files) so that its last digit is 1, and continue. See Chapter 30.

IY102 PART FILE NOT FOUND: Mount the correct disks on the appropriate drives and continue.

IY103 PART FILE MOUNTED ON WRONG DRIVE: See IY102.

IY104 UNEXPECTED END OF FILE ON PART FILE: The part file is null or the header record incorrect. Use the TYPE command to check for the former. The simplest recovery procedure is to bring your backup file up to date. See Chapter 30.

IY105 not used

IY106 RENAME FAILED: A rename failed during processing. There should be no damage to the file. See Chapter 30 for how to rename the file.

IY107 A QUOTE IS AN INVALID CHARACTER: No double quotes allowed in any file.

IY108 INVALID PART NUMBER: Must be A-Z, 0-9, a slash (/), or a dash (-).

IY109 NOT AN INTEGER: 0-9 only, no decimals or commas.

IY110 NOT NUMERIC: Not a numeric character, or too many digits to the left or to the right of the decimal. See documentation for size limits.

IY111 THAT'S NOT A VALID RESPONSE: You entered an invalid control character for that field. See documentation for appropriate response.

IY112 RENAME FAILED AT END: One or more item files will need to be renamed. There should be no damage to the file. See Chapter 30 for how to rename the file.

IY113 INVALID PART FILE KEY: For the access key @D(rive)N(umber), D is not a valid drive reference, or N a valid record number.

IY114 PART NUMBER NOT FOUND: Either the file is not in sorted order, or the part specified is not on the file.

IY115 A SORT WILL BE REQUIRED IF THIS PART IS ADDED: Only the file affected will need to be sorted. See Chapter 22 for sorting procedures.

IY116 PART NUMBER ALREADY IN FILE: No duplicate part numbers allowed.

IY117 NO PR2 FILE FOUND: Your system disk is not in the drive where it was at the start of the current run. If not, contact your distributor.

IY118 RECORD NUMBER OUT OF RANGE: For the access key @D(rive)N(umber), N is not a valid record number on drive D.

IY119 INVALID PART NUMBER: see IY108

IY120 NO PR2 FILE FOUND: Contact your distributor.

IY121 THERE ARE NO VALID PART RECORDS. YOU'LL HAVE TO ADD SOME: Should appear only at system start up.

IY122 THE PART RECORD IS BEING DELETED: informational message.

PARAMETER ENTRY (IYPARENT.INT)

IY201 THIS PROGRAM MUST BE RUN FROM THE IY MENU: Type CRUN2 IY to enter the Inventory Control System.

IY202 INVALID INPUT: Invalid control characters or an invalid response. Consult the documentation for the appropriate response.

IY203 not used

IY204 INVALID DRIVE SPECIFIED: System drive must be either A or B.

IY205 not used

IY206 NO PARAMETER FILE. WILL CREATE DEFAULT FILE BEFORE CONTINUING: This message should only appear at system start up. If you do receive it, stop the program and locate the parameter file, or allow the program to create one.

IY207 UNEXPECTED END OF PARAMETER FILE: Parameter file not found when stop requested. Contact your distributor.

ADD TO STOCK (ADDITION.INT)

IY301 PART FILE FOUND FROM ABORTED RUN: see IY101

IY302 PART FILE NOT FOUND: see IY102

IY303 PART FILE MOUNTED ON WRONG DRIVE: see IY102

IY304 UNEXPECTED END OF FILE ON PART FILE: see IY104

IY305 not used
IY306 not used
IY307 not used

IY308 PR2 FILE NOT FOUND: The program was unable to find the PR2 file, or found a null one. Either PIP a good PR2 file onto the system disk, or recreate one with the Parameter Entry program. This should only if disks are switched improperly during ITEM FILE RECOPY or sorting.

IY309 TOO MANY DIGITS: Your entry was not numeric, or contained too many digits. The fields for quantity added, quantity on order, or quantity on back order accept up to 6 digit entries.

IY310 INVALID RESPONSE: An inappropriate control character was entered. See the documentation for the valid response.

IY311 INVALID RESPONSE: re-enter appropriate data.

IY312 QUANTITY IN STOCK MAY NOT BE LESS THAN ZERO

IY313 CURRENT PERIOD STOCK ADDITIONS CANNOT GO BELOW ZERO

IY314 CURRENT PERIOD STOCK ADDITIONS CANNOT EXCEED 999999

IY315 TO DATE STOCK ADDITIONS CANNOT GO BELOW ZERO

IY316 TO DATE STOCK ADDITIONS CANNOT EXCEED 999999

IY317 CANNOT BE LESS THAN ZERO: The quantity on order or back-ordered cannot be less than zero or greater than 999999.

IY318 CANNOT EXCEED 999999: An attempt was made to either reduce an activity accumulator below zero or increase one beyond its maximum. The accumulator will be set to zero or its maximum by the program.

IY319 INVALID DATE: Dates must be of the form MM/DD/YY or DD/MM/YY, depending on what was specified in the Parameter Entry program.

IY320 AUDIT FILE CREATE FAILED: The ADD TO STOCK program was unable to create the audit file. Perhaps the disk is full, or is write-protected.

IY321 AUDIT FILE FOUND FROM PREVIOUS ABORTED RUN: The ADD TO STOCK or DEPLETE STOCK program ended abnormally on its previous run, leaving an audit file of type 000. Find and correct the cause of the aborted run. If the audit file appears to be complete, rename it to type 001 and continue, otherwise, erase it. See Chapter 30.

IY322 AUDIT FILE OPEN FAILED: The ADD TO STOCK program was unable to open an audit file which was listed in the directory, or found a null file. Erase the file and continue. See Chapter 30.

IY323 Y, N, OR ESCAPE ONLY: Re-enter appropriate data.

IY324 NOT NUMERIC: Re-enter appropriate data.

IY325 NULL AUDIT FILE FOUND: Possible hardware problem.

IY326 PART VALUE SET TO 9,999,999.99: This is the maximum inventory value allowed for an item. If it is not large enough for your needs, congratulations on your good fortune.

IY327 AUDIT FILE OPEN FAILED AFTER SAVING FILES: see IY322

IY328 INVALID PART NUMBER: The part number specified does not exist in the part file(s). Re-enter a valid part number: A-Z, 0-9, slashes (/), or dashes (-) only.

IY329 INVALID PART KEY: A part key must be of the form @Dnnn, where D is the drive reference for the desired part file, and nnn is the record number of the desired part within that file.

IY330 VALUE SET TO ZERO: The transaction would take the value below zero.

DEPLETE STOCK (DEPLETE.INT)

IY351 PART FILE FOUND FROM ABORTED RUN: see IY101

IY352 PART FILE NOT FOUND: see IY102

IY353 PART FILE MOUNTED ON WRONG DRIVE: see IY102

IY354 UNEXPECTED END OF FILE ON PART FILE: see IY104

IY355 INVALID DEPLETION QUANTITY: Your entry was not numeric, or its length was greater than 6.

IY356 PR2 FILE NOT FOUND: see IY308

IY357 not used

IY358 PR2 FILE NOT FOUND: see IY308

IY359 UNEXPECTED EOF ON AUDIT FILE: Contact your distributor.

IY360 INVALID RESPONSE: Valid entries include CTRL-N, CTRL-B, ESCAPE, and, if a valid record has been read, RETURN, CTRL-R, and CTRL-S.

IY361 PART VALUE SET TO MIN. (ZERO): The depletion based on the current average value per unit has forced the inventory value to zero.

IY362 STOCK MAY NOT BE LESS THAN ZERO: Your entry would set the quantity on hand below zero.

IY363 CURRENT DEPLETIONS AT MIN. (ZERO): informational message.

IY364 CURRENT DEPLETIONS AT MAX. (9,999,999): informational message.

IY365 TO DATE DEPLETIONS AT MIN. (ZERO): informational message.

IY366 TO DATE DEPLETIONS AT MAX. (9,999,999): informational message.

IY367 PART VALUE SET TO MAX. (9,999,999): When backing out a depletion, the inventory value has exceeded the maximum, based on the current average value.

IY368 STOCK MAY NOT EXCEED 999,999: Your entry would cause the stock on hand quantity to exceed the maximum.

IY369 INVALID RESPONSE: This field will accept only a valid depletion quantity, CTRL-B, CTRL-R, CTRL-X, or RETURN.

IY370 AUDIT FILE CREATE FAILED: Your disk or directory may be full. If not, contact your distributor.

IY371 AUDIT FILE FOUND FROM PREVIOUS ABORTED RUN: An IYAUD.000 file has been found, meaning the ADD TO STOCK or DEPLETE STOCK program ended abnormally. Determine the reason for the abnormal ending, rename the file to type 001, and continue. See Chapter 30.

IY372 AUDIT FILE OPEN FAILED: Contact your distributor.

IY373 Y, N, OR ESCAPE ONLY: Y, N, T, True, Ok, or ESCAPE only.

IY374 not used
IY375 not used
IY376 not used

IY377 OPEN FAILED: Contact your distributor.

IY378 INVALID PART NUMBER: See IY328.

IY379 INVALID PART KEY: See IY329.

START AN ACTIVITY PERIOD (UPDATE.INT)

IY401 PART FILE FOUND FROM PREVIOUS ABORTED RUN: see IY101

IY402 PART FILE NOT FOUND: see IY102

IY403 PART FILE MOUNTED ON WRONG DRIVE: see IY102

IY404 UNEXPECTED END OF FILE ON PART FILE: see IY104

IY405 PR2 FILE NOT FOUND OR NULL: see IY308

IY406 ACTIVITY REPORT HAS NOT BEEN RUN: Run the Activity Report, if desired, and continue.

IY407 INVALID RESPONSE: Inappropriate entry. See the documentation for the proper response.

IY408 Y, YES, N, OR NO ONLY: Re-enter appropriate data.

IY409 not used

IY MENU (IY.INT)

IY421 NO PARAMETER FILE: Contact your distributor.

IY422 DMS ERROR: CRT.DEF is an invalid definition file. Contact your distributor.

IY423 ERROR DURING CHAIN TO SORT: Disk or directory full, or disk is write protected. See Chapter 30. If none of these, contact your distributor.

IY424 ERROR DURING CHAIN TO SORT: see IY423

IY425 THE PROGRAM IS NOT ON THE CURRENTLY LOGGED DISK: All INT files should be located on the currently logged drive.

IY426 THE SORT PARAMETER FILE IS NOT ON THE SYSTEM DISK: Run the CREATE SORT PARAMETER FILES function.

IY427 ERROR DURING CHAIN TO SORT: see IY423

IY428 not used

IY429 THIS PROGRAM CAN ONLY BE RUN ON VERIFIED SORTED PART FILES: All item files must be sorted and checked for duplicate records (by running the ITEM FILE PRINT program) before the program selected can be run.

IY430 INVALID SECOND PARAMETER FILE: Invalid information in the IYPR2 parameter file. See Chapter 30.

IY431 QSORT IS NOT ON THE CURRENTLY LOGGED DISK: QSORT is expected on the currently logged disk. See documentation Chapter 22 for sorting procedures.

IY432 THAT'S NOT A VALID RESPONSE: The menu selection must be numeric. ESC or CTRL-R (Refresh the screen) are the only other valid commands.

IY433 NO SECOND PARAMETER FILE. CHAINING TO PARTFILE: Since no parameter file exists, the program will create one and allow you to change the default values. PARTFILE is the program responsible for the ITEM FILE RECOPY function.

IY434 THE SYSTEM DRIVE HAS NOT BEEN INITIALIZED: You cannot continue until you specify either drive A or drive B as the system drive (the drive where the parameter and SRT files are expected).

IY435 INVALID DATE: DD/MM/YY or MM/DD/YY only.

IY436 INVALID PARAMETER FILE: IYPR1 contains invalid information. See Chapter 30.

IY437 OPERATOR REQUESTED RETURN TO MENU: informational message.

IY438 **ERROR** THE PROGRAM WAS TERMINATED BEFORE COMPLETION: A fatal error occurred in the program just run. Determine the cause, if possible, and continue. See Chapter 30.

IY439 TRANSACTION HISTORY IS NOT CURRENTLY RECORDED: Auditability was not requested. Run PARAMETER ENTRY program to select it if desired.

IY440 not used

REORDER REPORT (REORDER.INT)

IY441 PART FILE MOUNTED ON WRONG DRIVE: see IY103

IY442 UNEXPECTED EOF ON PART FILE: The part file was not found or contains invalid information. Find the partfile, or bring your backup file up to date.

IY443 THAT'S NOT A VALID RESPONSE: Only numeric entries allowed, no control characters.

IY444 INVALID PART NUMBER: see IY108

ITEM FILE PRINT (PARTPRNT.INT)

IY461 PART FILE FOUND FROM PREVIOUS ABORTED RUN: see IY101

IY462 PART FILE NOT FOUND: see IY102

IY463 PART FILE MOUNTED ON WRONG DRIVE: see IY102

IY464 UNEXPECTED END OF FILE ON PART FILE: see IY104

IY465 PART FILE RENAME FAILED: The program tried to rename an item file of type 111 to 101 at end.

IY466 DUPLICATE PART NUMBER FOUND ON DRIVE: RUN PARTENT TO DELETE: Programs in the Inventory Control System will not operate properly if there are duplicate part numbers. Delete the flagged duplicates with the ITEM ENTRY program, and re-run the ITEM FILE PRINT program.

IY467 not used

IY468 PR2 FILE NOT FOUND: see IY308

IY469 INVALID RESPONSE: Re-enter appropriate data.

IY470 INVALID PART NUMBER: A-Z, 0-9, slash (/), or dash (-) only.

IY471 TYPE 111 PART FILE FOUND WHEN TYPE 101 PRESENT: Remove 101 file.

CREATE SORT PARAMETER FILES (SORTPARM.INT)

IY491 INVALID RESPONSE: Inappropriate entry. See documentation for proper response.

IY492 DRIVE REFERENCE MUST BE A-Z: Re-enter appropriate data.

IY493 UNABLE TO CREATE: Directory may be full or disk may not be write enabled.

IY494 UNABLE TO WRITE: The CREATE SORT PARAMETER FILES program was unable to create or write to the specified file. Perhaps the disk is full or write-protected.

VALUATION REPORT (VALUE.INT)

IY501 INVALID PART NUMBER: see IY108

IY502 STARTING PART NUMBER GREATER THAN ENDING PART NUMBER: Re-enter appropriate data.

IY503 PART FILE MOUNTED ON WRONG DRIVE: see IY102

IY504 UNEXPECTED END OF FILE ON PART FILE: see IY104

IY505 not used

IY506 INVALID RESPONSE: Re-enter appropriate data.

ACTIVITY REPORT (ACTIVITY.INT)

IY511 PART FILE ON WRONG DRIVE: See IY102.

IY512 UNEXPECTED EOF ON PART FILE: See IY104.

IY513 PROGRAM MUST BE RUN FROM MENU: See IY201.

IY514 OPTION NUMBER OUT OF RANGE: Your entry was not numeric, greater than 6, or less than 1.

IY515 INVALID PART NUMBER: A-Z, 0-9, slash (/), or dash (-) only. A RETURN defaults to the first part number on the file.

IY516 INVALID PART NUMBER: A-Z, 0-9, slash (/), or dash (-) only. A RETURN defaults to the end of the file.

IY517 ENDING PART NUMBER IS LESS THAN STARTING PART NUMBER

IY518 PR2 FILE NOT FOUND: See IY308.

TRANSACTION PROOF (PROOF.INT)

IY521 PROGRAM MUST BE CALLED FROM MENU: See IY201.

IY522 PROGRAM CAN'T RUN. NO AUDITING REQUESTED: Change the audit parameters through the PARAMETER ENTRY program if auditability desired.

ITEM FILE RECOPY (PARTFILE)

IY531 UNEXPECTED EOF ON PART FILE READ: Invalid information in part file header record. See Chapter 30.

IY532 INVALID RESPONSE: Not a valid response. See documentation for valid responses.

IY533 INVALID PART FILE PARAMETERS: See documentation for proper file allocation.

IY534 END OF FILE ON PART FILE CREATE: See Chapter 30.

IY535 THAT'S NOT THE RIGHT INPUT FILE: Put the correct file on the input drive.

IY536 NO INPUT FILE FOUND ON DRIVE A: Put the correct item file on drive A.

IY537 INVALID PART FILE: The input file is null. See Chapter 30.

IY538 THAT'S NOT THE RIGHT OUTPUT FILE: You've changed the B disk before being prompted to do so.

IY539 UNEXPECTED EOF ON WRITE TO PART FILE: See IY423 and Chapter 30.

IY540 NO OUTPUT FILE FOUND: You've removed the B disk before being prompted.

IY541 INVALID OUTPUT FILE FOUND: The output file is null. See Chapter 30.

IY542 THERE IS A PART FILE ON THE DISK ON DRIVE B: No part files should reside on the disk used to receive the recopied output. Mount a disk that does not contain an IYPRT file.

IY543 THERE IS NO SECOND PARAMETER FILE ON THAT DISK: Check to make sure your system disk (the disk with IYPR1 and IYPR2) is in the drive named as the system drive (either A or B).

IY544 UNABLE TO CREATE PART FILE. IS THE DISK WRITE PROTECTED?: Same as IY423. See Chapter 30.

IY545 UNABLE TO WRITE TO PART FILE: see IY544.

IY546 INVALID DATE: MM/DD/YY or DD/MM/YY only.

IY547 UNABLE TO CREATE PR2 FILE: see IY544.

IY548 NO INPUT FILE FOUND ON DRIVE A: Put item file to be recopied on drive A as directed.

IY549 INVALID PART FILE: The file on the input drive is null. See Chapter 30.

IY550 INVALID PART FILE: Files must be in sorted order. If copying continues, item files will not be in sorted order.

30. Error Recovery

When certain programs, notably the data and parameter entry, recopy, and update programs end abnormally they may leave a file with an improper file type. Under normal conditions the file types of item, parameter, and audit files should end with a 1 (one) to indicate the file is a valid current file (for example, IYAUD.001). Since the 1 is changed to a 0 (zero) when the file is being processed, and renamed back to 1 when processing is complete, you will never see a zero as the third digit unless the program ended before the rename took place, or the rename failed for some other reason.

Before renaming the file with the CP/M REN command, you should try to learn why the program ended prematurely or the rename failed. It is quite possible that the problem is hardware related, and it may even be due to a problem with the operating system.

To rename a file, give the REN command in the following form:

```
REN D:nnnnnnnn.ttt=D:nnnnnnnn.ttt
```

where D is a drive reference, nnnnnnnn an 8 character file name, and ttt the 3 character file type. The file name and type to the left of the equal sign is that to which the file on the right will be renamed. For example, to rename IYPRT.100 to IYPRT.101 give the command,

```
REN A:IYPRT.101=IYPRT.100
```

if IYPRT.100 is on A, and you want IYPRT.101 on the A drive as well. The rename cannot take place if a file with the same name as that to which the other file is to be renamed already exists on the designated drive.

Certain error conditions may arise because a disk or its directory is full, or the disk write-protected. If the disk is write protected, remove the write-protect tab or apply a write-enable tab (whichever is appropriate for your system). Unless your disk contains many other files besides those used and created by the Inventory Control System, the disk directory should not be full. However, since the number of files a directory can hold does vary, you should find out the capacity of the directory you are using.

To check to see if a disk is full use the CP/M STAT command. Although this command should be on the currently logged drive, it can give you the status of files on other drives. For example, to find the number of bytes remaining on the B drive with STAT on the A drive type the command,

```
STAT B:
```

See the CP/M Features and Facilities manual for complete instructions on how to use the STAT command.

Some error messages make reference to a null file or null header record. An INVALID FILE error message may also indicate a null file since no SSG files are created as null files. All SSG files have one record written to them upon creation, either a header record, or if a parameter file, the entire file is made up of one record. If the entire file is null, the TYPE

command, instead of displaying the file contents, will simply return the CP/M prompt character. If the header record is null, the first record to appear will be a data record. Currently only the item and audit files use header records.

If a header record is incorrect, you may be able to correct it using a file editor. Check the file specifications for the necessary information. Be certain to get the record length correct. This recovery procedure is suggested only for those with substantial operating experience or data processing knowledge. In any case, back up your files before attempting to modify them with an editor. For others, the best solution is to bring your backup file up to date.

To recover from certain error conditions you may need to recreate the first of the two parameter files IYPR1. Since no information in this file is altered by processing, once the file has been created by the PARAMETER ENTRY program you can continue.

If a recovery procedure requires you to recreate the second parameter file IYPR2 you may need to alter the file to get back to where you were. This is because the IYPR2 file contains information that is altered during the normal processing cycle. You may want to reset the batch number to correspond with the current number. To do so, check the file specifications and make the changes with an editor. The IYPR2 file also contains a flag which indicates whether any files are unsorted. You need not reset this flag if you make sure all your item files are in sorted order before continuing.

Message IY438 **ERROR** THE PROGRAM WAS TERMINATED BEFORE COMPLETION appears at the system menu after a fatal error. The error that caused IY438 to appear should have displayed its error message briefly before the menu and IY438 appeared. If for some reason you did not notice which error it was, restart the system and try to continue where you left off. A rename may be required if the program just run was either ADD TO STOCK, DEPLETE STOCK, ITEM ENTRY, OR START A NEW ACTIVITY PERIOD. If the error occurs again, be sure to take note of the message.

31. Command Summary

ESCAPE: Ends the current function or program.

RETURN: Enters data and advances to next request, accepts default value, moves cursor forward through DMS screen fields.

STOP: Typing S-T-O-P followed by RETURN is the same as ESCAPE, use at non-DMS screen requests only.

CTRL-H: Backs cursor up one position, erasing the character in that position.

DELETE (SHIFT-DELETE): Backs up cursor one position, erasing the character in that position; use at non-DMS requests.

CTRL-R: Refreshes garbled DMS screen.

CTRL-X: Cancels current add to stock, deplete stock, or item entry transaction.

CTRL-A: At ITEM ENTRY access key field, puts program in add-records mode.

CTRL-D: At ITEM ENTRY access key field, deletes record currently displayed.

CTRL-S: At ITEM ENTRY access key field, saves records added or changed so far this session.

CTRL-N: At ITEM ENTRY access key field, if no record currently displayed, gets first record on file; if a record is currently displayed, gets next record after it.

CTRL-B: At ITEM ENTRY access key field, gets record before currently displayed record; within a DMS screen, moves cursor back one field.

^ (UP ARROW): Moves cursor back one request at non-DMS screen request.

32. Inventory Specifications Manual

Inventory Control System
Specifications Manual

Inventory Control System

Specifications Manual

September 1979
Copyright (c) 1979 - 2025.

SSG, Applewood Computers and Vincent B Coen

I. Distribution Disk Programs

PARTENT.INT
IYPARENT.INT
ADDITION.INT
DEPLETE.INT
UPDATE.INT
IY.INT
REORDER.INT
PARTPRNT.INT
VALUE.INT
ACTIVITY.INT
PROOF.INT
PARTFILE.INT

CRTDEF.INT

SDCOPY.COM
QSORT.COM

ADM3A.DEF
B150.DEF
B152.DEF
B157.DEF
CRO3100.DEF
DYNABYTE.DEF
SOROC120.DEF
VIOC.DEF

II. File Accessing

ADD AND DEPLETE

FILES READ:	FILES WRITTEN TO:
IYPRT.100	IYPR2.101 IYPRT.100 (1 to 3) update in place IYAUD.001

FILES CREATED:	WHEN:
IYAUD.000	if no audit file present or if requested

FILES RENAMED:		WHEN:
from:	to:	
IYPRT.101	IYPRT.100	at beginning (1 to 3)
IYAUD.001	IYAUD.000	at beginning if not created
IYPRT.100	IYPRT.101	at end (1 to 3)
IYAUD.000	IYAUD.001	at end

PARTENT

FILES READ:	FILES WRITTEN TO:
IYPRT.100*	IYPRT.110 if present
IYPRT.110*	IYPRT.100 if present
IYPRT.000*	IYPRT.000 if present
*one file of any of these three types on 1 to 3 parameter specified drives	IYPR2.101 at end of processing

FILES CREATED:	WHEN:
none	

FILES RENAMED:		WHEN:
from:	to:	
IYPRT.111	IYPRT.110	at startup, if present
IYPRT.101	IYPRT.100	at startup, if present
IYPRT.001	IYPRT.000	at startup, if present
IYPRT.110	IYPRT.111	at end, if present
IYPRT.100	IYPRT.101	at end, if present
IYPRT.000	IYPRT.001	at end, if presnet

FILES DELETED:	WHEN:
none	

PART FILE

FILES READ:	FILES WRITTEN TO:
IYPRT.101 when recopying	IYPRT.100 when recopying
	IYPR2.101 at end of processing

FILES CREATED:	WHEN:
IYPRT.100	when recopying
IYPRT.101	at system startup
IYPR2.101	if not present

FILES RENAMED:		WHEN:
from:	to:	
IYPRT.100	IYPRT.101	at end of processing

FILES READ:	FILES WRITTEN TO:
IYPR1.101 when first run	$$$.SUB
IYPR2.102 when first run	
CRT.DEF if present when first run	

FILES CREATED:	WHEN:
$$$.SUB	if sort requested

FILES RENAMED:	WHEN:
none	

FILES DELETED:	WHEN:
none	

REORDER

FILES READ:	FILES WRITTEN TO:
IYPRT.101 on up to 3 parameter specified drives	none

FILES CREATED:	WHEN:
none	

FILES RENAMED:	WHEN:
none	

FILES DELETED:	WHEN:
none	

IYPARENT

FILES READ:	FILES WRITTEN TO:
IYPR1.101 at start	IYPR1.101 at end

FILES CREATED:	WHEN:
IYPR1.101	if none present

FILES RENAMED:	WHEN:
none	

FILES DELETED:	WHEN:
none	

PARTPRNT

FILES READ:	FILES WRITTEN TO:
IYPRT.001 or 101 or 111 (1 to 3)	IYPR2

FILES CREATED:	WHEN:
none	

FILES RENAMED:		WHEN:
from:	to:	
IYPRT.111 (1 to 3)	IYPRT.101	if no duplicates

FILES DELETED:	WHEN:
none (at present)	

UPDATE

FILES READ:	FILES WRITTEN TO:
IYPRT.100 (1 to 3)	IYPRT.100 (1 to 3)

FILES CREATED:	WHEN:
none	

FILES RENAMED:		WHEN:
from:	to:	
IYPRT.101	IYPRT.100	at beginning (1 to 3)
IYPRT.100	IYPRT.101	at end (1 to 3)

FILES DELETED:	WHEN:
none	

SORTPARM

.sp3
.li

FILES READ:	FILES WRITTEN TO:
	PART1.SRT
	PART2.SRT if 2 item files
	PART3.SRT if 3 item files

FILES CREATED:	WHEN:
PART1.SRT	
PART2.SRT if 2 item files	
PART3.SRT if 3 item files	

FILES RENAMED:	WHEN:
none	

FILES DELETED:
none

VALUE

FILES READ:	FILES WRITTEN TO:
IYPRT.101 (1 to 3, according to parameter file)	none

FILES CREATED:	WHEN:
none	

FILES RENAMED:	WHEN:
none	

FILES DELETED:
none

III. File Descriptions

IYPRT (DATA RECORD)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
PRT.NUMBER$	10	3	2	Part number-file sort key
PRT.VENDOR$	5	3	14	Vendor for part
PRT.ORDER.DATE$	6	3	22	Date order placed
PRT.ORDER.DUE$	6	3	31	Date order due
PRT.DESC$	30/var	3	40	Description
PRT.CONSTRUCT	6/var	1	varies	not used
PRT.DELETED%	2/var	1	"	T/F delete flag
PRT.RETAIL	8/var	1	"	Retail price
PRT.VALUE	10/var	1	"	Inventory value
PRT.COST	8/var	1	"	Unit cost
PRT.REORD.PT	6/var	1	"	Reorder point
PRT.STD.REORD	6/var	1	"	Standard reorder quantity
PRT.BK.ORDER	6/var	1	"	Quantity back ordered
PRT.ON.ORDER	6/var	1	"	Quantity on order
PRT.STOCK	6/var	1	"	Quantity in stock
PRT.WIP	6/var	1	"	not used
PRT.STK.ADD	6/var	1	"	Level 1 accumulator for additions
PRT.STK.DEP	6/var	1	"	Level 1 accumulator for depletions
PRT.TD.STK.ADD	7/var	1	"	Level 2 accumulator for additions
PRT.TD.STK.DEP	7/var	1	"	Level 2 accumulator for depletions
PRT.WIP.ADD	6/var	1	"	not used
PRT.WIP.DEP	6/var	1	"	not used
PRT.TD.WIP.ADD	7/var	1	"	not used
PRT.TD.WIP.DEP	7/var	0	"	not used
cr/lf		2		

TOTAL RECORD LENGTH : 214

IYPRT (HEADER RECORD)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
PRT.NO.RECS%	5/var	1	varies	Number of records in the file
PRT.DELETED.RECS%	5/var	1	"	Number of deleted records
PRT.LAST.UPDATE$	6	3	"	Date file last updated
PRT.DRIVE$	1	3	"	Drive that file must run on
PRT.HIGH.PRT.NO$	10/var	2	"	Highest part number in file

IYPR1 (PARAMETER FILE #1)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
PR1.CO.NAME$	60/var	3	varies	Company name
PR1.DEBUGGING%	2/var	1	"	T/F If true, enables print of information useful to programmers
PR1.CRT.DEF.FILE$	8/var	3	varies	currently unused
PR1.BELL.USED%	2/var	1	"	T/F If true, console bell sounds on detecting an error
PR1.MANU.USED%	1/var	1	varies	not used
PR1.OE.USED%	1/var	1	"	not used
PR1.INVOICE.USED%	1/var	1	"	not used
PR1.PURCHASING.USED%	1/var	1	"	not used
PR1.ANY.AUDIT.USED%	2/var	1	"	T/F If true, at least one type of audit trail is currently used
PR1.ADDITIONS.AUDIT.USED%	2/var	1	varies	T/F If true, additions to stock are audited
PR1.DEPLETIONS.AUDIT.USED%	2/var	1	varies	T/F If true, depletions of stock are audited
PR1.START.JOB.AUDIT.USED%	1/var	1	varies	not used
PR1.START.JOB.AUDIT.SUMMARY.USED%	1/var	1	"	not used
PR1.END.JOB.AUDIT.USED%	1/var	1	"	not used
PR1.END.JOB.AUDIT.SUMMARY.USED%	1/var	1	"	not used
PR1.AUDIT.DRIVE$	1	3	"	Disk drive to place audit file
PR1.BOM.DRIVE$	1	3	"	not used
PR1.CURRENT.PERIOD$	7/var	3	"	Prints in report headings
PR1.TO.DATE.PERIOD$	7/var	3	"	Prints in report headings
PR1.BOM.INDICATOR$	0/var	3	"	not used
PR1.ARRAY.SIZE%	4/var	1	"	Array size used by heuristic binary search routine
PR1.LINES.PER.PAGE%	3/var	1	varies	Number of print lines on 1 page
PR1.PAGE.LENGTH%	3/var	1	"	Physical page length in lines (one-sixth of an inch per line
PR1.PAGE.WIDTH%	3/var	1	"	Page width to center report headings
PR1.DATE.MO%	1/var	1	"	Position of
PR1.DATE.DY%	1/var	1	"	month,day,yr
PR1.DATE.YR%	1/var	1	"	in external date, currently supports MM/DD/YY, DD/MM/YY (1,2,3 or 2,1,3 respectively)
PR1.64.COL.CRT%	1/var	1	varies	not used
PR1.AVERAGE.USED%	2/var	1	"	T/F If true, averaged valuation is computed
PR1.1%,PR1.2%,PR1.3%,PR1.4%,PR1.5%, PR1.6%,PR1.7%,PR1.8%,PR1.9%,PR1.10%, PR1.11%,PR1.12%,	1/var	1	varies	Spares
PR1.S1$,PR1.S2$,PR1.S3$	0/var	3	"	Spares
PR1.S4$	0/var	2	"	Spares

IYPR2 (PARAMETER FILE #2)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
PR2.AUDIT.NUMBER%	2/var	1	varies	Current audit batch number
PR2.ACTIVITY.REPORT.RUN%	2/var	1	"	T/F If true, the activity report has been run for all parts
PR2.PART.FILE.SORT.NEEDED%	2/var	1	varies	T/F If true, at least one part file must be sorted before regular processing can continue
PR2.PART.FILES%	2/var	1	varies	Number of system part files
PR2.PART.DRIVE$(1)	1	3	"	Drive to place 1st part file
PR2.PART.DRIVE$(2)	1	3	"	Drive to place 2nd part file
PR2.PART.DRIVE$(3)	1	3	"	Drive to place 3rd part file
PR2.LO.NUMBER$(2)	10/var	3	"	Lowest part number to be in part file #2
PR2.LO NUMBER$(3)	10/var	3	varies	Lowest part number to be in part file #3
PR2.LAST.UPDATE$	6	3	varies	Date of last part file update
PR2.1%,PR2.2%,PR2.3%, PR2.4%,PR2.5%	1/var	1	"	Spares
PR2.S1$	0/var	3	"	Spares
PR2.S2$	0/var	2	"	Spares

IYAUD (AUDIT FILE)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
AUD.PRT.NO$	10	3	2	Part number
AUD.DESC$	30/var	3	14	Part description
AUD.DEPLETION%	1/var	1	varies	not used
AUD.REVERSE.TRANS%	2/var	1	"	T/F True if a transaction cancellation
AUD.TRANS.QTY	6/var	1	"	Transaction quantity
AUD.VALUE.CHG	11/var	1	"	Inventory value change
AUD.UNIT.COST	8/var	0	"	Unit cost

TOTAL RECORD LENGTH : 80

IYAUD (HEADER RECORD)

FIELD NAME	LENGTH	OVHD	START	FMT/COMMENTS
AUD.PROOFED%	2/var	1	varies	T/F True if proofed since last updated
AUD.PROOF.NO%	5/var	1	"	Number of proofs run on current batch
AUD.ADD.RUN%	2/var	1	"	T/F True if an additions batch
AUD.DEP.RUN%	2/var	1	"	T/F True if a depletion batch
AUD.START.JOB%	1/var	1	"	not used
AUD.END.JOB%	1/var	1	"	not used
AUD.RUN.DATE$	6	3	"	Date last modified
AUD.NO.RECS%	5/var	0	"	Number of audit records

End of IY Manual